SM-X916B/X916C

Service Schematic

TAB_S9_ULTRA_REV0.3

sheet01 : RF (1/2)

sheet02 : RF (2/2), RF PMIC

sheet03 : TRCV (SDR735 x 2)

sheet04 : BT / WIFI(WCN7851), GPS

sheet05 : AP_SM8550(Secondary PCB), JTAG, UFS

sheet06 : AP_SM8550(Secondary PCB)

sheet07 : PMIC(Secondary PCB), IF PMIC(CC IC), D/C

sheet08 : AUDIO

sheet09 : CAMERA / DISPLAY

sheet10 : SENSOR / SD_SIM / KEY / IF

sheet11 : WACOM / ANT SUB PCB

sheet12 : USB PCB

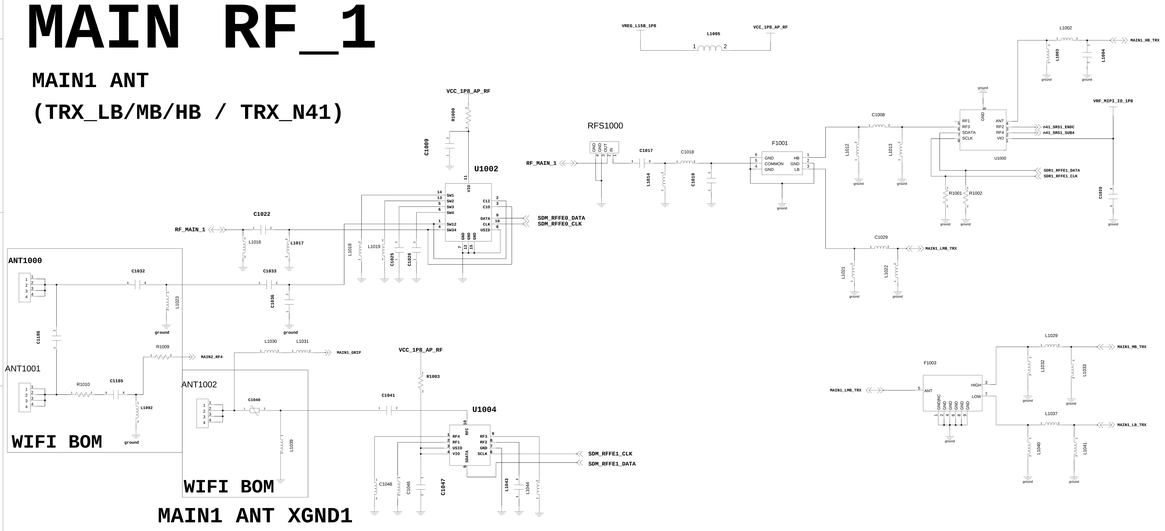
sheet13 : WIFI TR/TL PCB

sheet14 : Secondary PCB PAD

TITLE	TabS7P		
SIZE A1	DWG NO	PDM NUMBER	REV Rev
DRAWN BY User Name	SHEET 0 of 14	Date Drawing Date	

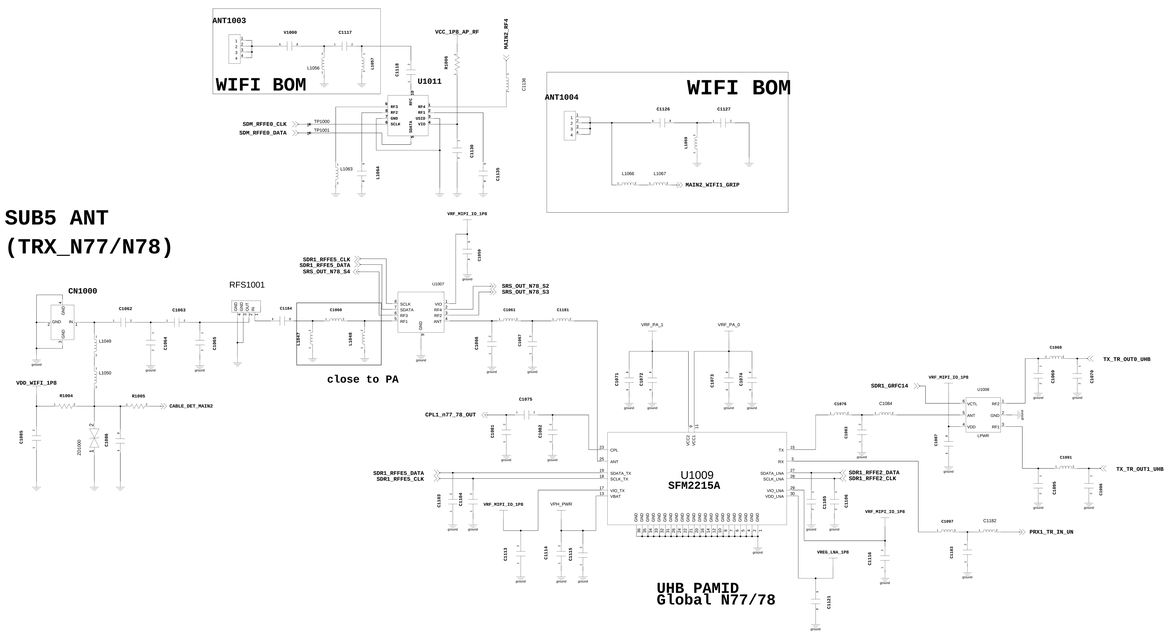
MAIN RF_1

MAIN1 ANT
(TRX_LB/MB/HB / TRX_N41)

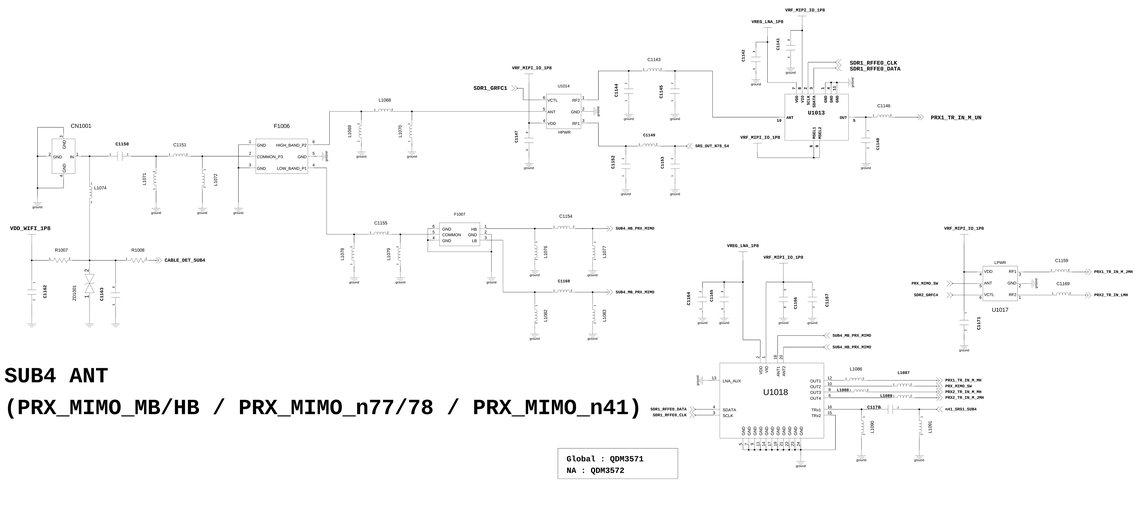


MAIN1 ANT XGND1

SUB5 ANT
(TRX_N77/N78)

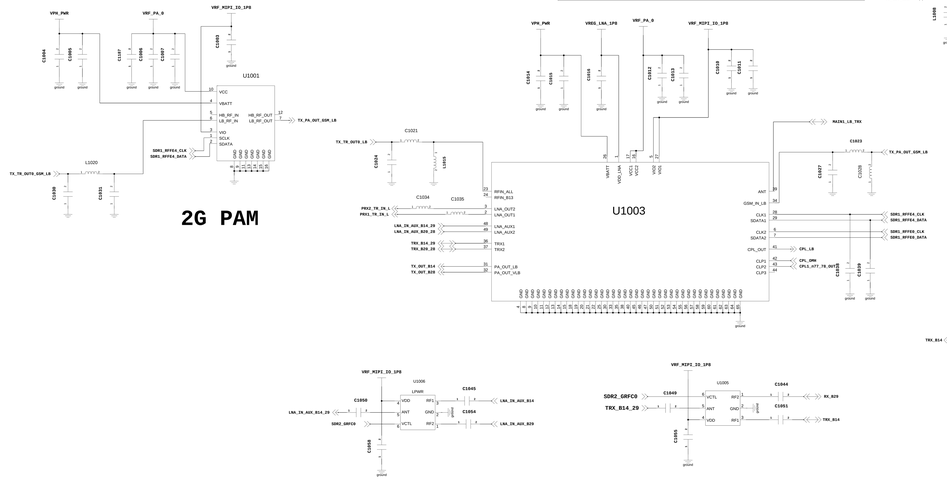


SUB4 ANT
(PRX_MIMO_MB/HB / PRX_MIMO_n77/78 / PRX_MIMO_n41)

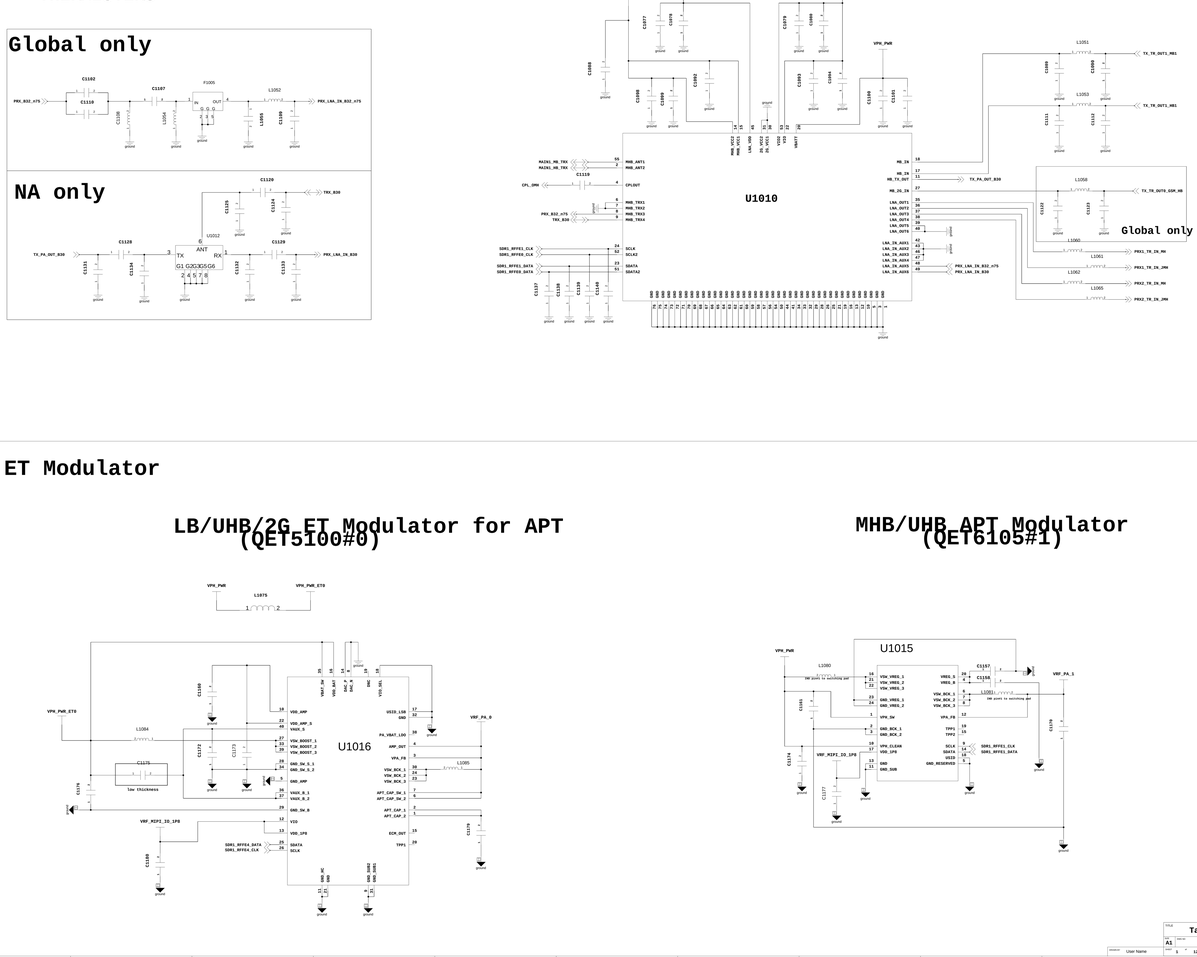


Global : QDM3571
NA : QDM3572

LB LPAMID
Global B5/8/12/13/17/20/26/28, G850/900



OMH LPAMID
Global 1/2/3/4/7/25/32/34/38/39/40/41/66, G1800/1900, N41



ET Modulator

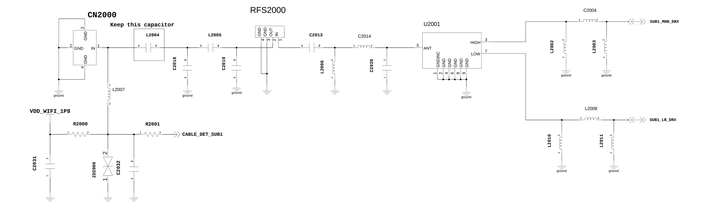
LB/UHB/2G ET Modulator for APT (QET5100#0)

MHB/UHB APT Modulator (QET6105#1)

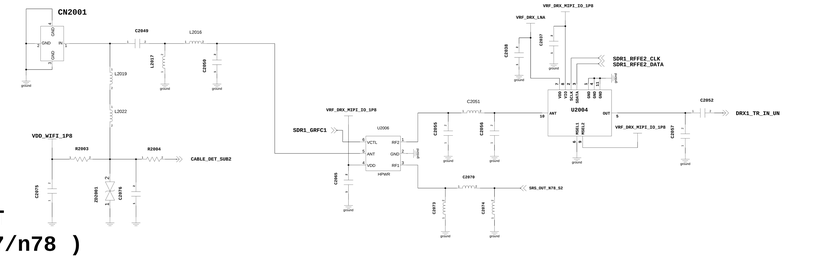
21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

TabS7P
A1
PDM NUMBER

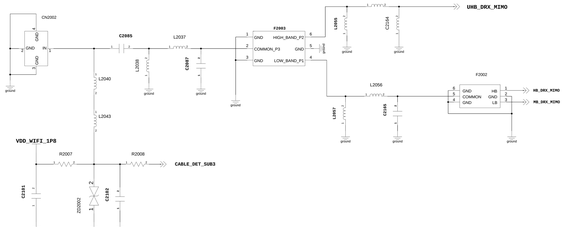
MAIN RF_2



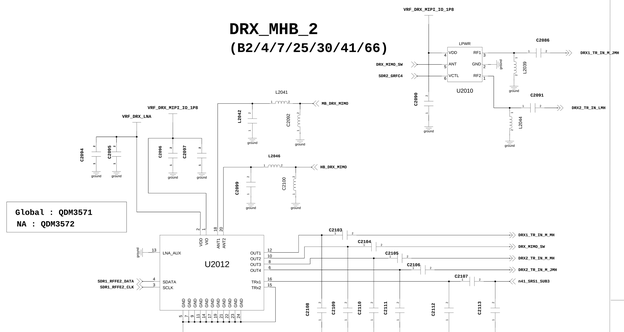
SUB1 ANT
(DRX_LB/MB/HB, DRX_n41, ENDC_LB/MB/HB TX)



SUB2 ANT
(DRX_n77/n78)

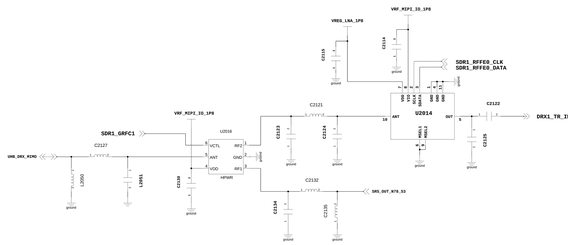


DRX_MHB_2
(B2/4/7/25/30/41/66)

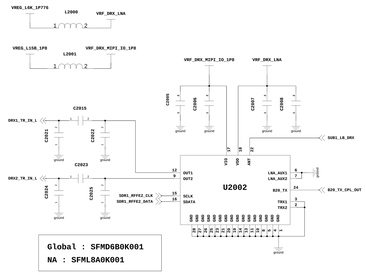


Global : QDM3571
NA : QDM3572

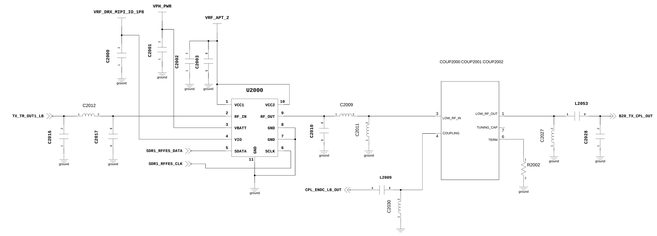
SUB3 ANT
(DRX_MIMO_MB/HB, DRX_MIMO_n41, DRX_MIMO_N77/78)



DRX_LB
(B5/8/12/13/14/17/20/26/28/29/71_G850/900)

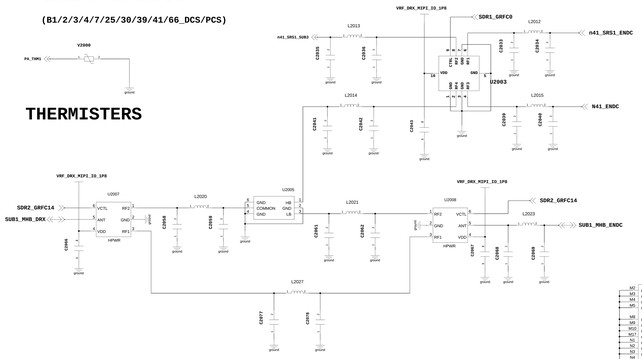


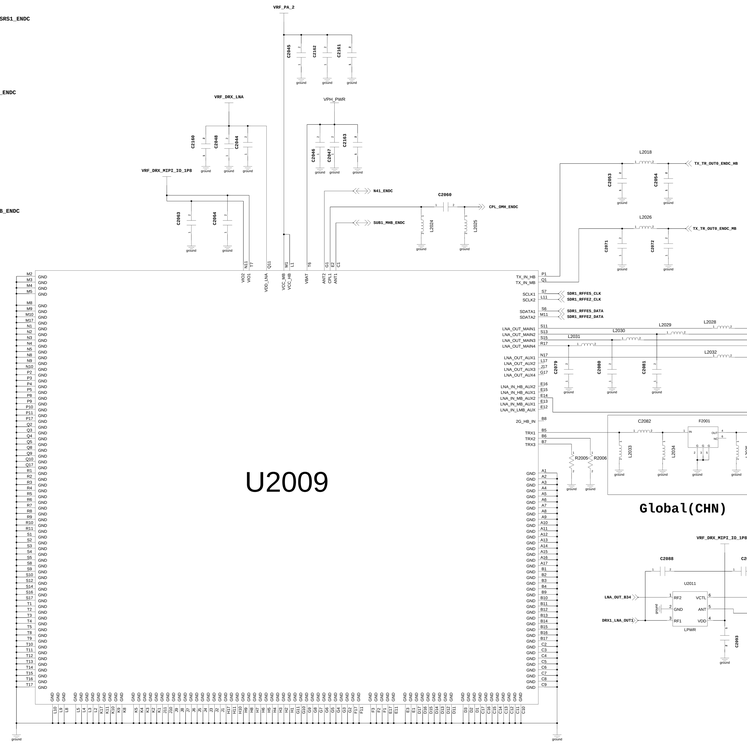
Global : SFMD6B0K001
NA : SFMLBA0K001



ENDC module

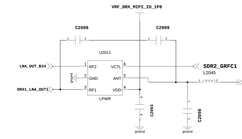
(B1/2/3/4/7/25/30/39/41/66_DCS/PCS)



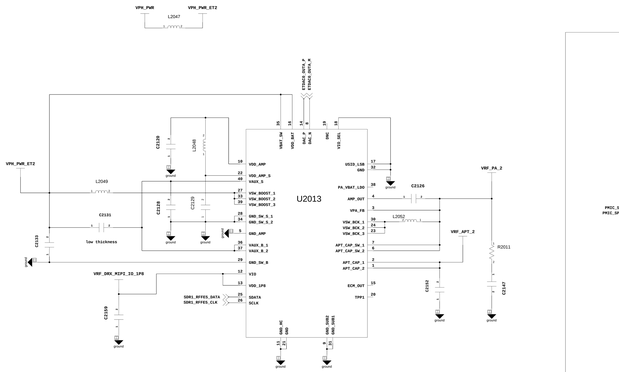


U2009

Global(CHN)

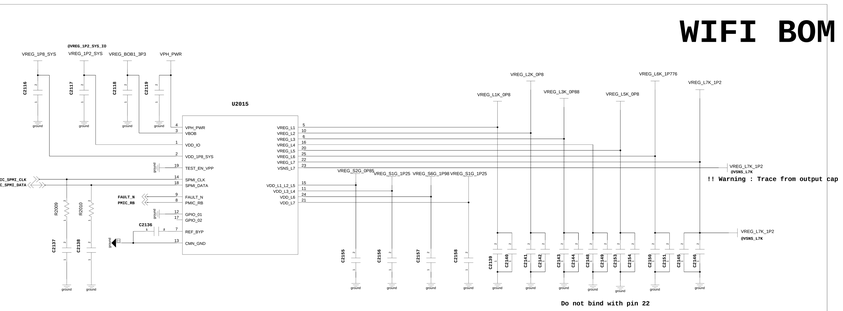


ENDC ET Modulator
(QET5100#2)



RF PMIC

WIFI BOM



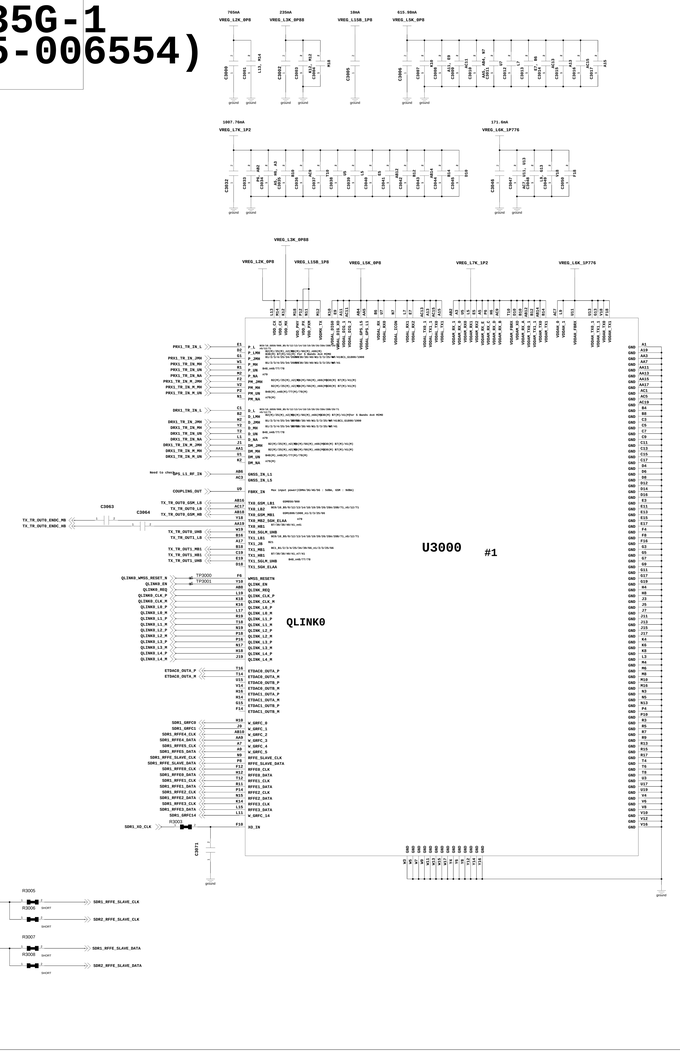
PMR735D #0

Title: TabS7P			
Size: A1	Number: PCBA_NUMBER		Rev:
Author: User Name	Date:	Sheet: 9 of 9	Drawing Name:

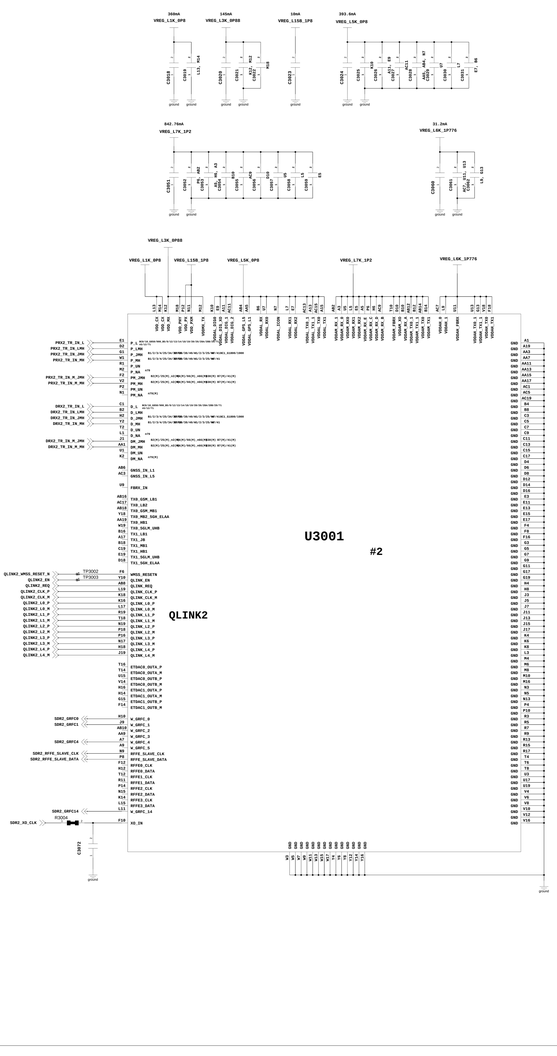
MAIN RF_3

TRANSCEIVER(Dual SDR735)

WIFI : SDR735G-1 (1205-006554)



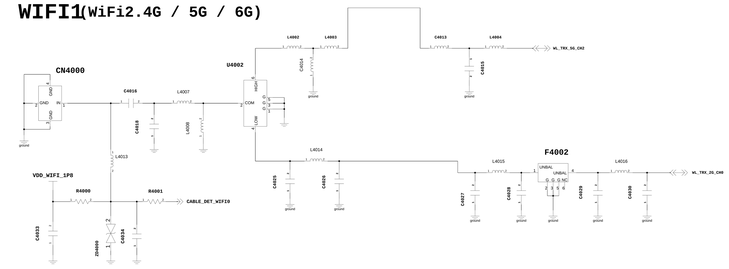
WIFI : NC

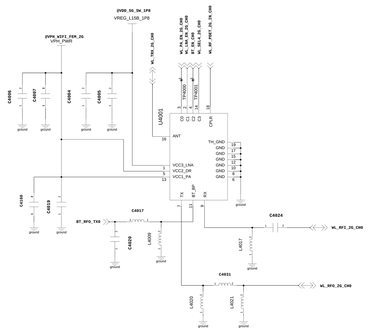


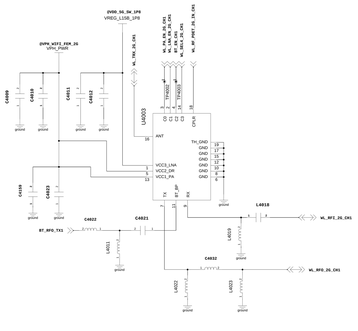
TITLE	TabS7P
SIZE	A1
DRAWING NUMBER	Rev
APPROVED	User Name
SHEET	3 OF 16
DATE	Drawing Date

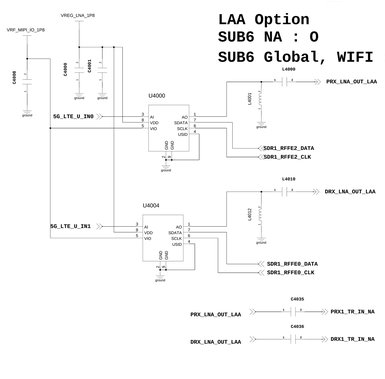
Connectivity

WIFI1 (WiFi2.4G / 5G / 6G)

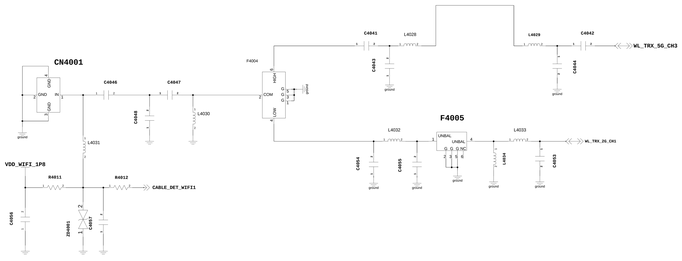


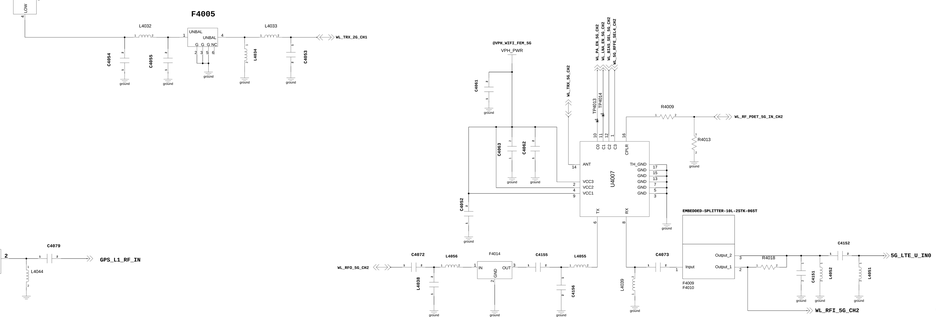


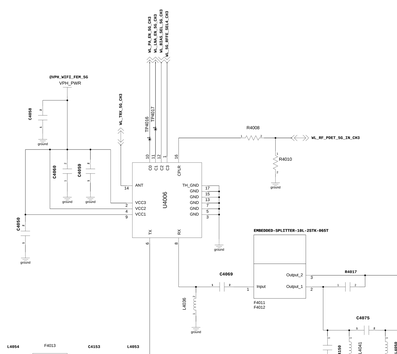




WIFI2 (GPS L1 / WiFi2.4G / 5G / 6G)



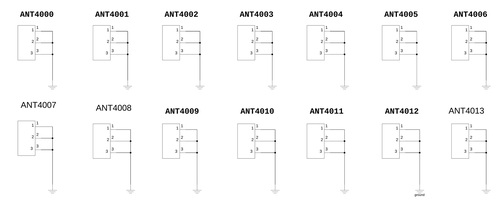




GPS

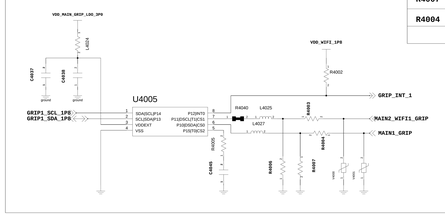
U4010

REAR CONTACT

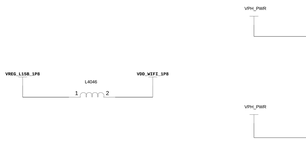


MAIN_GRIP_SENSOR(GRIP #2_#3)

	Sub6	WIFI
L4027	O	X
R4007	O	X
R4004	O	X



MAIN GRIP LDO



SM8550
(INTERFACE/GPIO)

System

CORE

on SECONDARY PCB

System

0802

REGION

ANT PCB	SHORT
X916(B/N)	RS100
X910(W)	RS101
X916(C)	RS102
ETC (E)	RS103

RS100 — B/N
RS101 — W
RS102 — C
RS103 — E

DRAM_POP (S_PCB)

12GB	1105-003212

UFS Memory (M_PCB)

128GB(UFS3.1)	1107- 002751
256GB(UFS4.0)	1107- 002905
512GB(UFS4.0)	1107- 002919
1TB(UFS4.0)	1107- 002924

0804
RS505 — 128GB
RS506 — 256GB
RS507 — 512GB
1TB

UFS 4.0 256G

BOM NC

AP JTAG CONNECTOR

SECURE NVM

CON5000
CON5001
2D BARCODE
SMD NC

JTAG5000

CAUTION: PLACEMENT
Near to TP AP_USB_BOOT

JTAG5001

JTAG5002

PCB array JTAG

WIFI NC

eUSB REPEATER

SIM Level Shifter

SD Level Shifter

CC I2C Option

TITLE	TabS7P
SIZE	A1
NUMBER	PDM NUMBER
REV	Rev
SHEET	1
OF	15
Drawing Date	
User Name	

SM8550 Power

(POWER, GND)

CORE

on SECONDARY PCB

21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1		

TITLE: Tab S7+

SIZE: A1 | DOCUMENT NUMBER: FORM NUMBER | REV: Rev

UNTOUCHED: User Name | SHEET: 6 OF 14 | DATE: Drawing Date

P PMIC set & CLK Generator

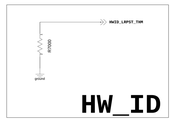
PMIC (PM8550/8550VS/8550VE)

!! CAUTION !!
Place Cin 10uF Bulk Caps at corner of PM8350 and as close to PM8350 as possible to achieve this requirement.

CORE

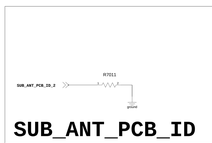
on SECONDARY PCB

Power System



H/W ID table (PM8550 AMUX_4)

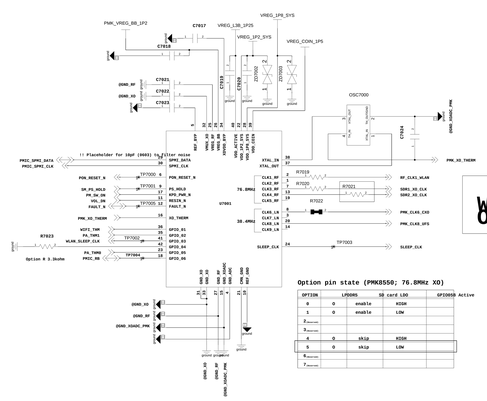
HW ID	BOARD REVISION	Value	Mech
00	Tab S9 Ultra REV0.0	6.04K	(HHP0543)
01		9.1K	
02		12.4K	
03	Tab S9 Ultra REV0.1	18K	2007-007135
04	Tab S9 Ultra REV0.2	22K	2007-007311
05	Tab S9 Ultra REV0.3	30K	2007-008275
06		39K	2007-007134
07		47K	2007-007139
08		56K	
09		68K	2007-007589
10		82K	2007-007590
11		100K	2007-007107
12		120K	2007-000163
13		150K	2007-007489
14		180K	
15		200K	
16			



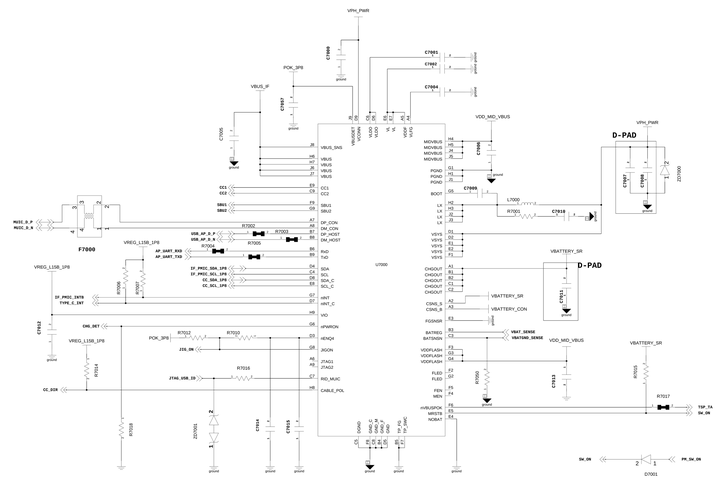
SUB ANT PBA Table

ID	ID_2	ID_1	ID_0
0	GND	GND	GND Global_Sub6
1	GND	GND	OPEN WIFI
2	GND	OPEN	GND Reserve
3	GND	OPEN	OPEN Reserve

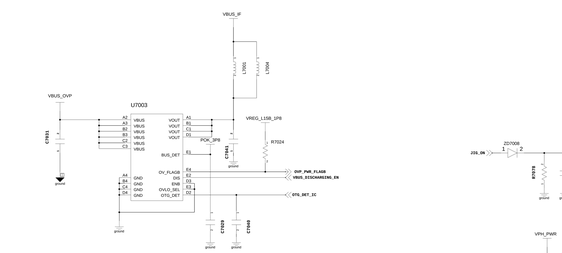
M_RF

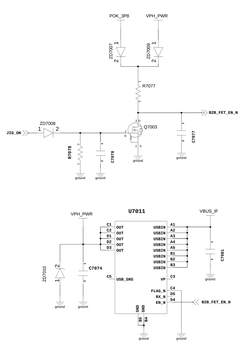


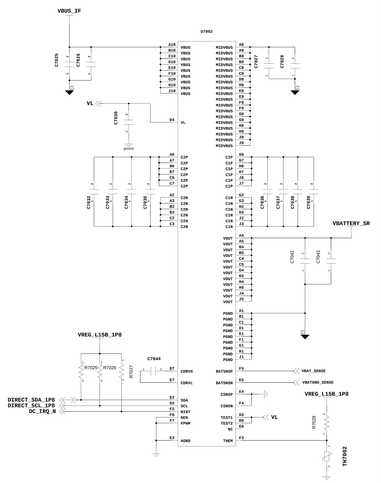
Power

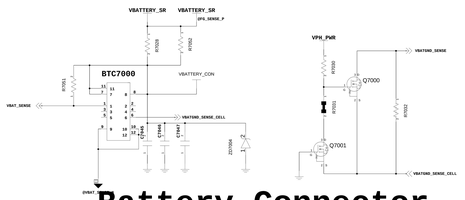


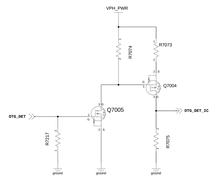
Power

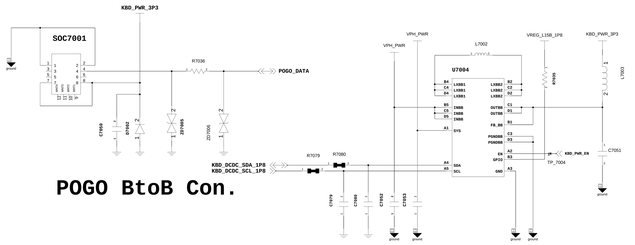


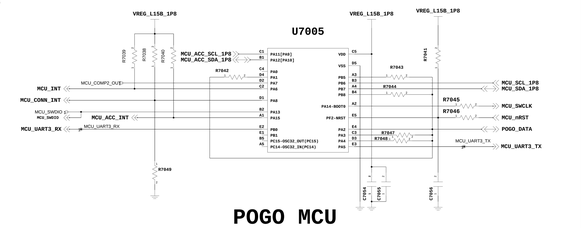


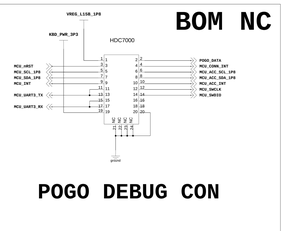












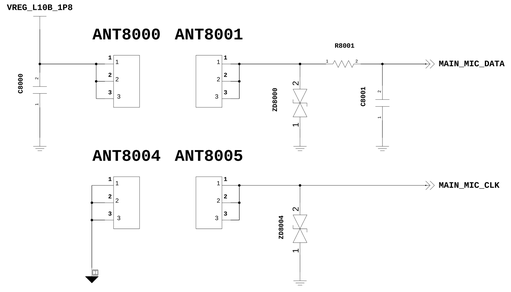
BOM NC

POGO

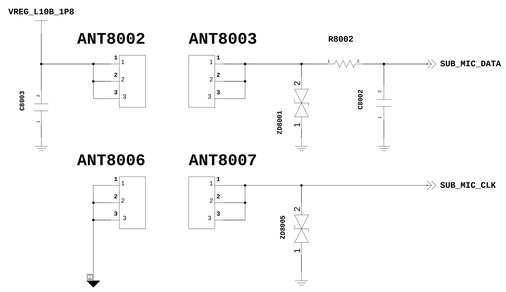
21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

MODEL NAME
PDM NUMBER
Rev
A1
User Name
Drawing Date

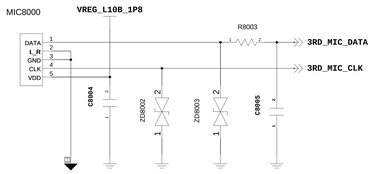
AUDIO



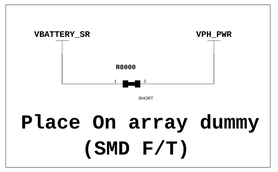
MAIN MIC

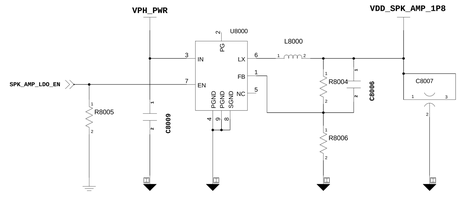


SUB MIC

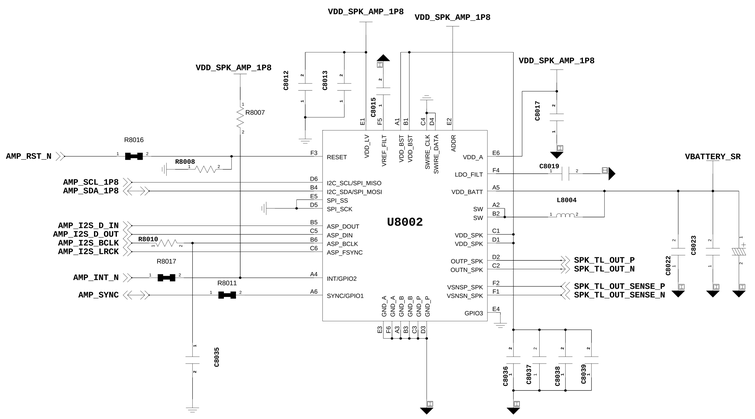


3rd MIC

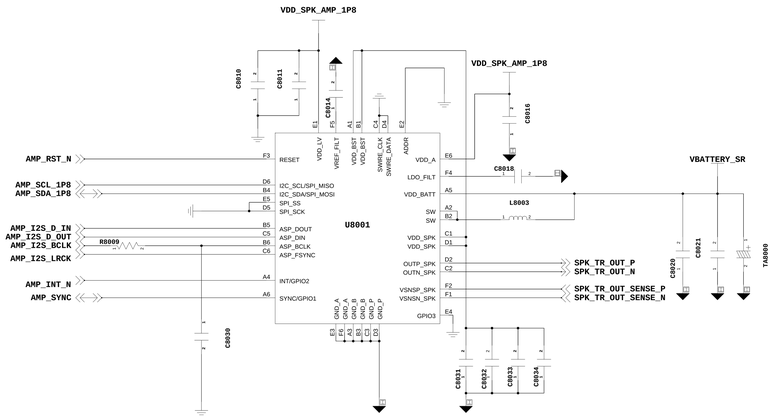




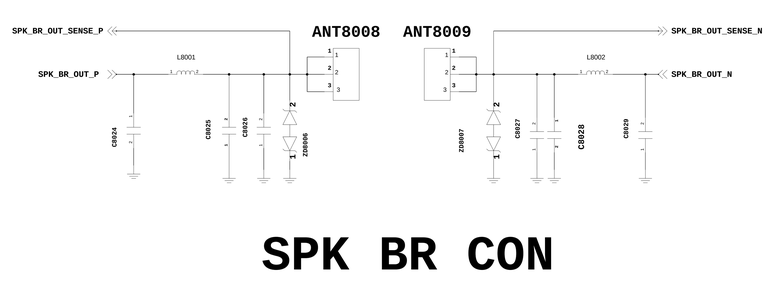
AUDIO DCDC



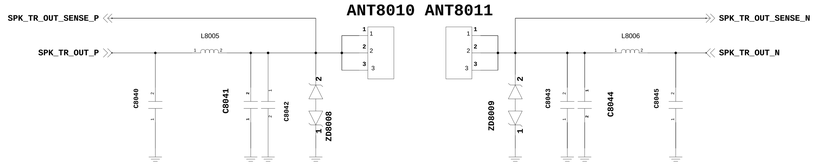
SPK TOP LEFT AMP



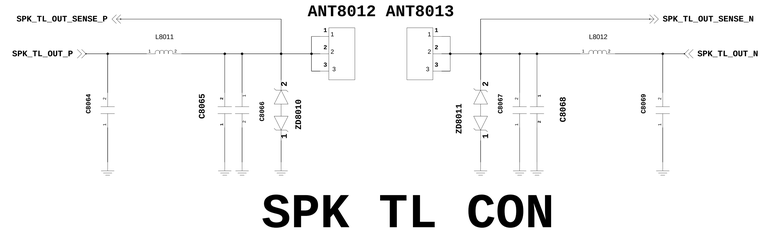
SPK TOP RIGHT AMP



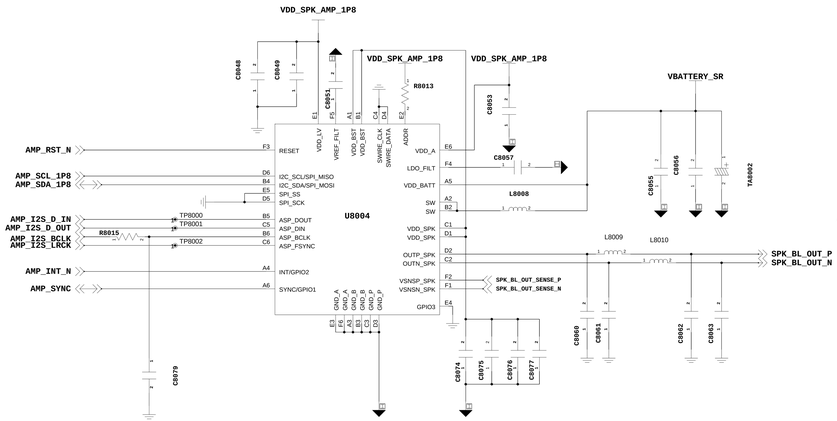
SPK BR CON



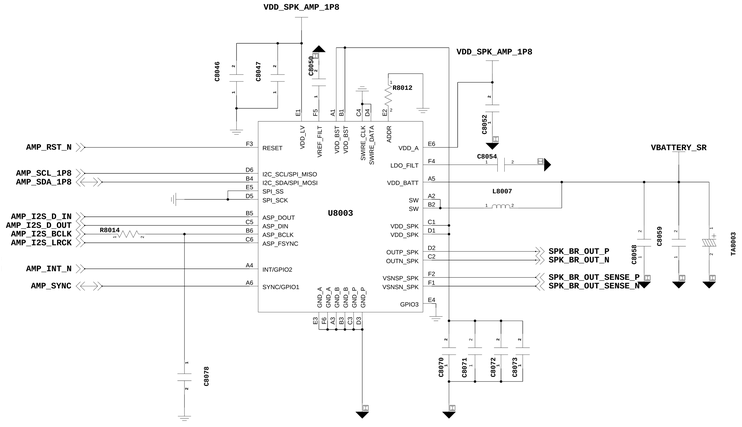
SPK TR CON



SPK TL CON



SPK BOTTOM LEFT AMP

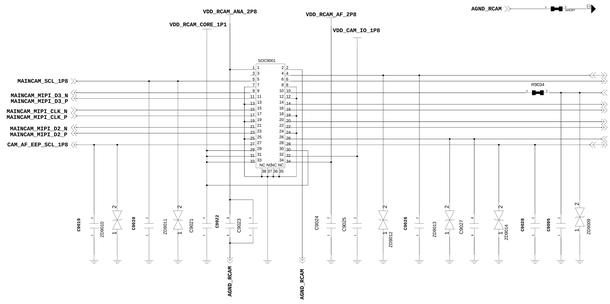


SPK BOTTOM RIGHT AMP

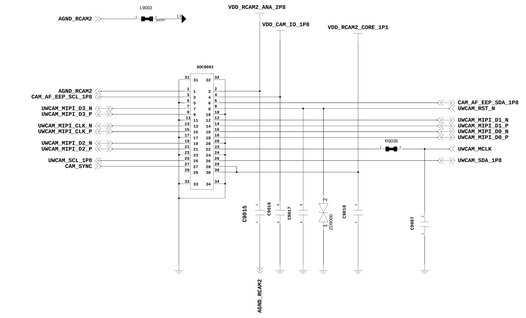
SPK TL CON

TITLE	TabS7P
SIZE	A1
DWG NO	PDM NUMBER
REV	Rev
DRAWN BY	User Name
SHEET	8 of 16
DATE	Drawing Date

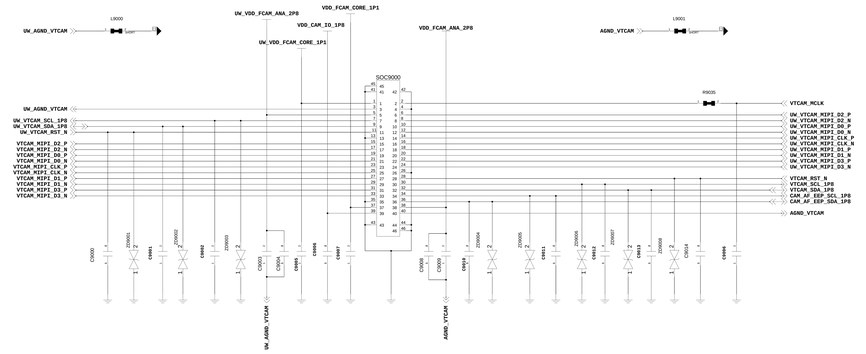
CAMERA



13M_CAM (RCAM)

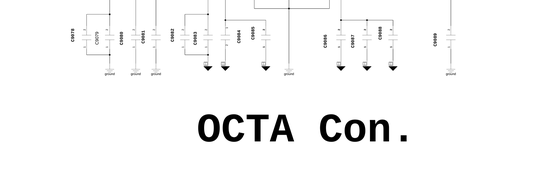


8M_UW CAM (RCAM2)

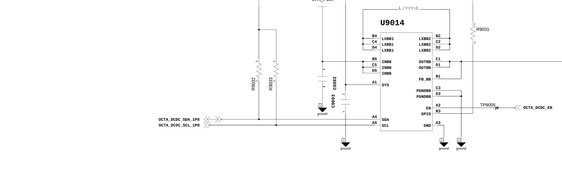


12M CAM (FCAM)

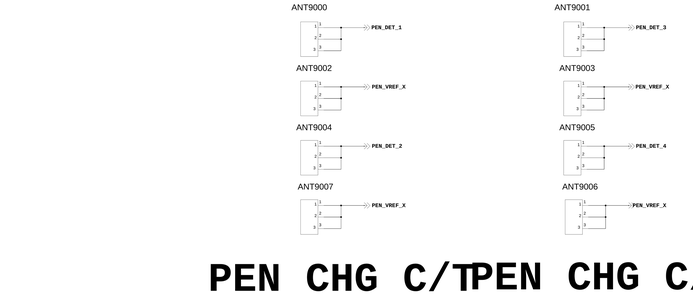
12M_UW CAM (FCAM)



13M_CAM (RCAM)_LDO



8M_UW CAM (RCAM2)_LDO

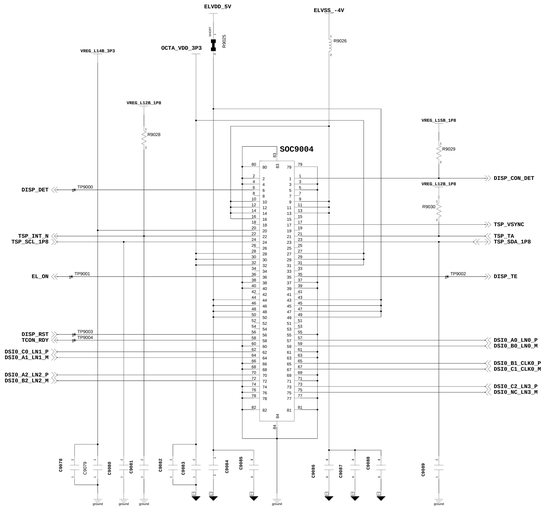


VTCAM_LDO

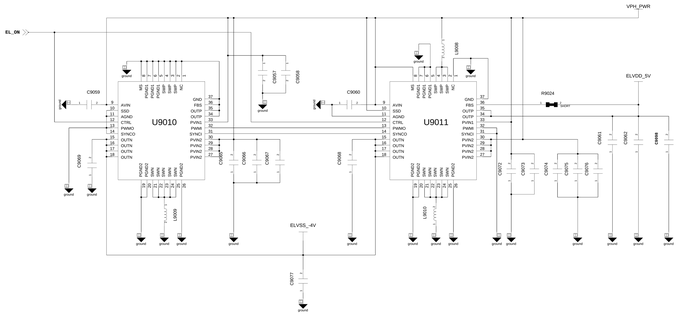
UW VT CAM_LDO

FLASH LED

DISPLAY

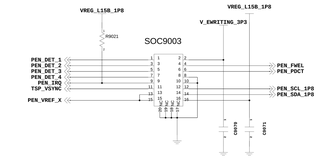


OCTA Con.



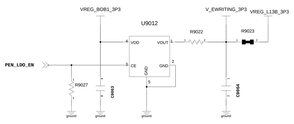
OCTA POWER

TCON BUCK BOOSTER



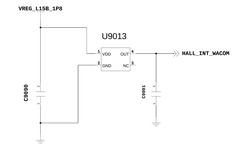
WACOM CONN

(MAIN <-> OCTA / WACOM)



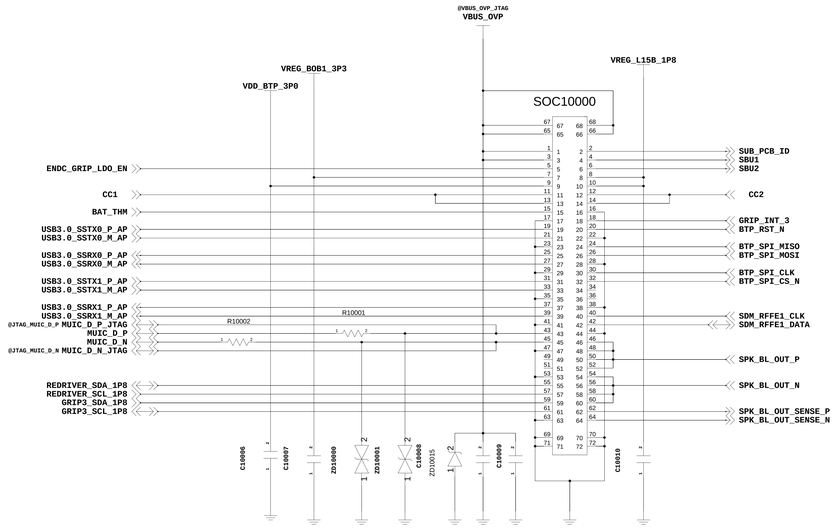
PEN LDO

PEN CHG C/TPEN CHG C/T_2

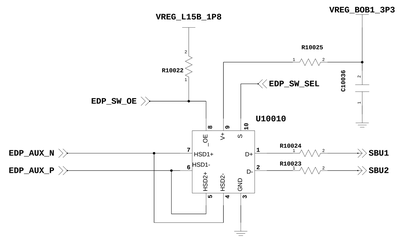


COVER_DET HALL_IC

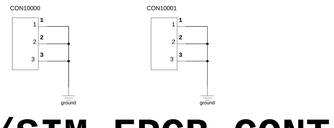
SENSOR / SD/SIM / KEY / I/F



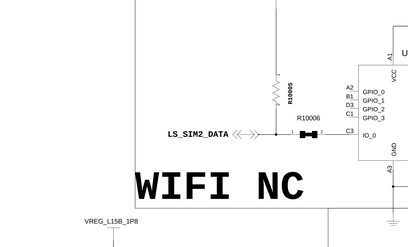
I/F CONN (MAIN <-> USB)



DP Switch



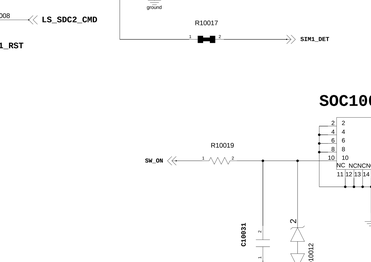
RGB Sensor 1



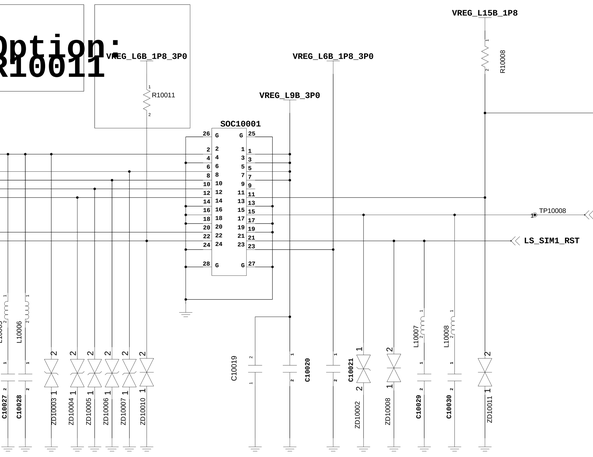
6 Axis (GYRO,ACCEL)

RGB Sensor 2

Option

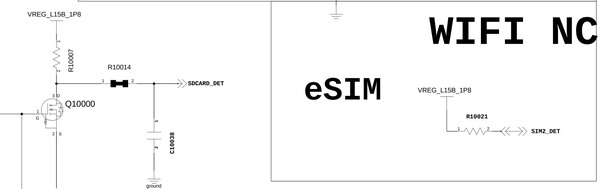


Magnetic SENSOR

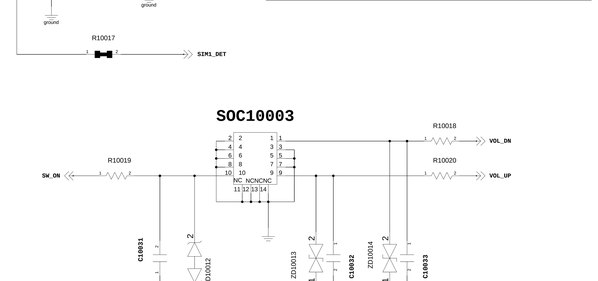


WIFI Option:
Open R10011

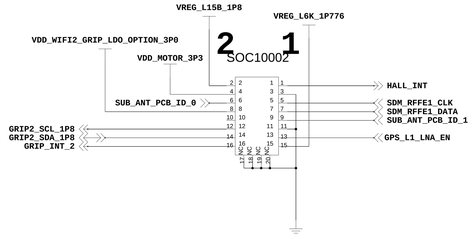
SIM/SD SOCKET Conn.



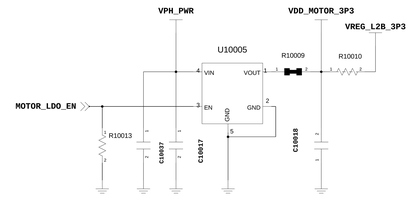
WIFI NC



WIFI NC
eSIM



SUB CONN (MAIN <-> ANT SUB)



MOTOR LDO

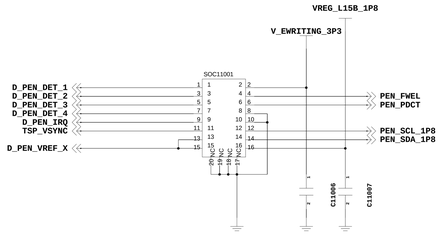
FOD LDO

SD/SIM FPCB CONTACT

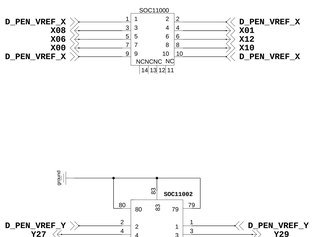
Key CONN

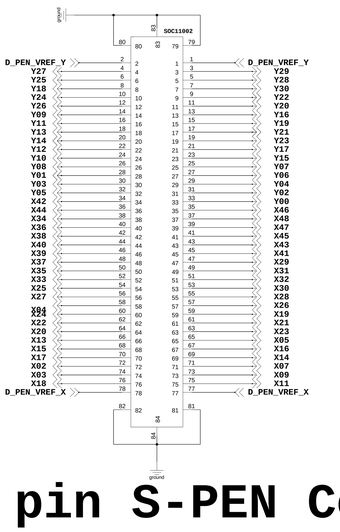
TITLE	TabS7P
SIZE A1	DWG NO PDM NUMBER Rev
drawn by User Name	SHEET 10 of 15 Date Drawing Date

WACOM PCB

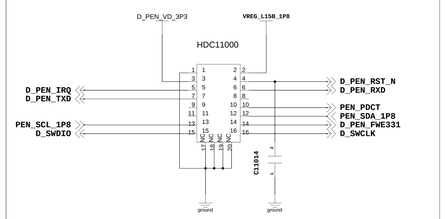


WACOM I/F CON.

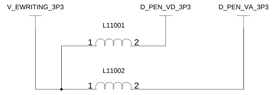




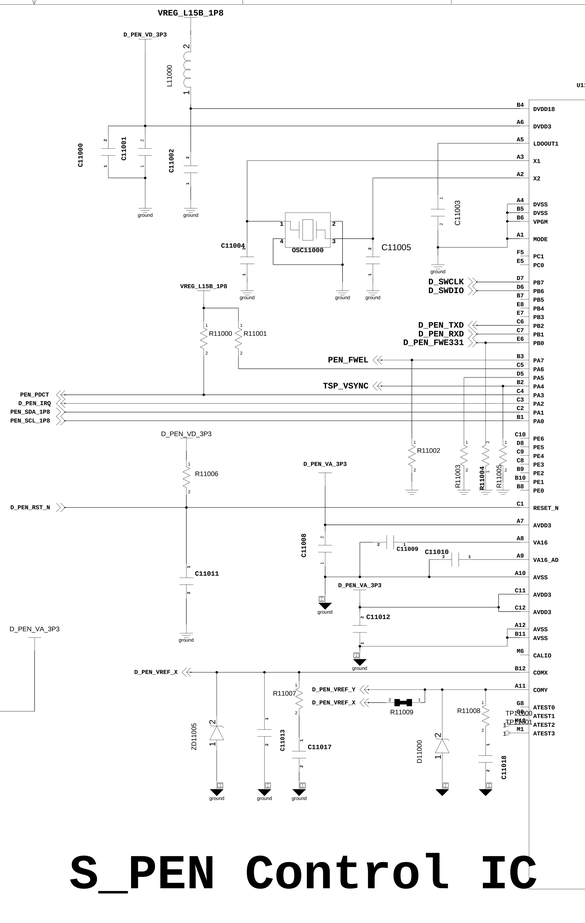
MBOM : NC



DEBUGER



78+10 pin S-PEN Connector



S_PEN Control IC

(DC mode)

< SMD F/T >

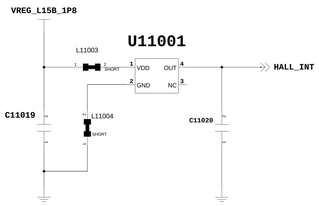
1. V_EWRITING_3P3
2. VREG_L15B_1P8
3. PEN_SCL_1P8
4. PEN_SDA_1P8
5. PEN_FWEL
6. PEN_PDCT
7. GND

ANT SUB PCB

(MAIN1 ANT XGND / WIFI2 ANT GRIP)

SCREW SUS

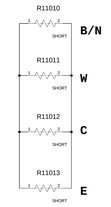
Sub6	WIFI
R11023	R11021
R11024	R11022
	RFS11001



COVER_DET HALL_IC

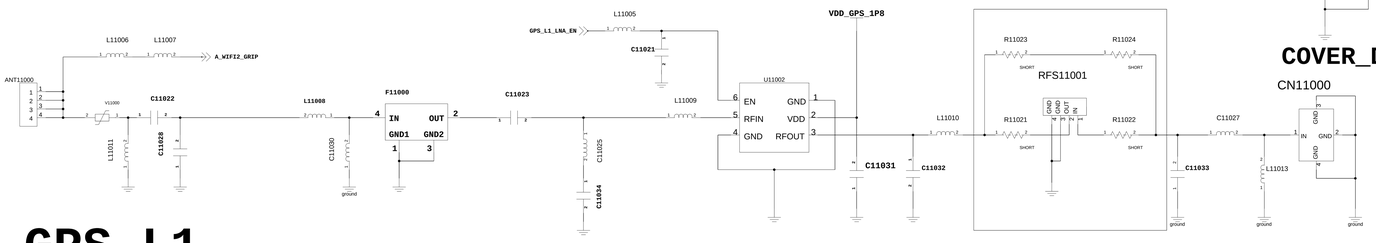
ANT PCB REGION

ANT PCB	SHORT
X916B(B/N)	R11010
X910(W)	R11011
X916(C)	R11012
ETC(E)	R11013

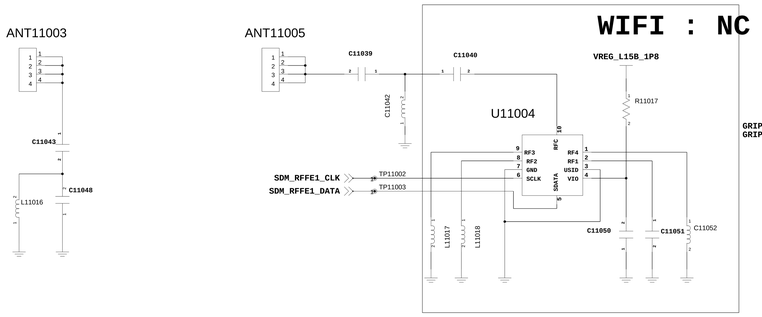


< SMD F/T >

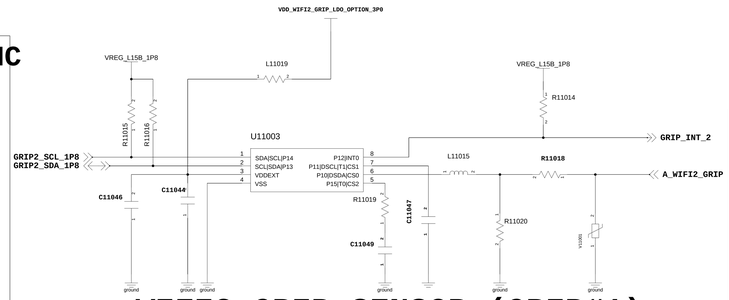
1. VDD_WIFI2_GRIP_OPTION_LDO_3P0
2. GRIP2_SCL_1P8
3. GRIP2_SDA_1P8
4. GRIP_INT_2
5. SDM_RFFE1_CLK
6. SDM_RFFE1_DATA
7. VREG_L15B_1P8
8. HALL_INT
9. GPS_L1_LNA_EN
10. VREG_L6K_1P776
11. GND
12. VPH_PWR
13. GRIP2_LDO_EN



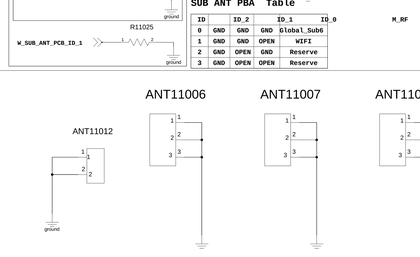
GPS_L1



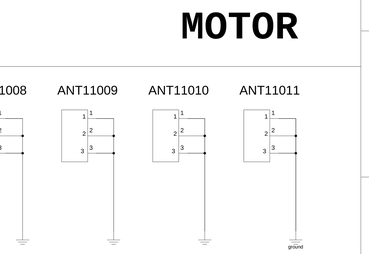
WIFI : NC



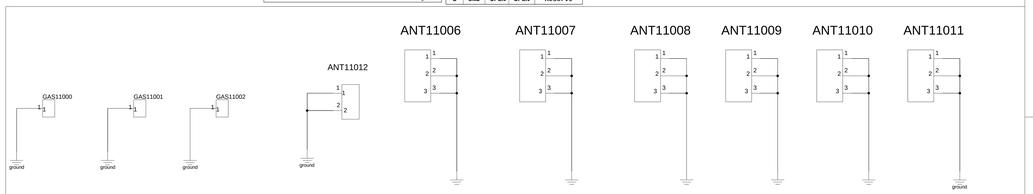
WIFI2_GRIP_SENSOR (GRIP#1)



ID	ID_2		ID_1	ID_0
0	GND	GND	GND	Global_Sub6
1	GND	GND	OPEN	WIFI
2	GND	OPEN	GND	Reserve
3	GND	OPEN	OPEN	Reserve



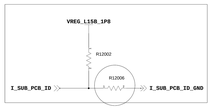
MOTOR



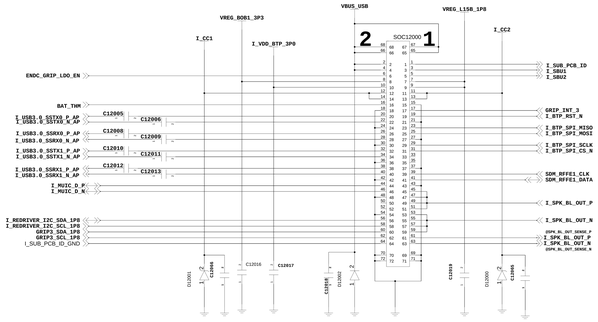
Front CONTACT Cable CLIP REAR CONTACT

TITLE	TabS7P		
SIZE A1	DWG NO PDM NUMBER		REV Rev
DRAWN BY User Name	SHEET 12 OF 14	DATE Drawing Date	

USB PCB

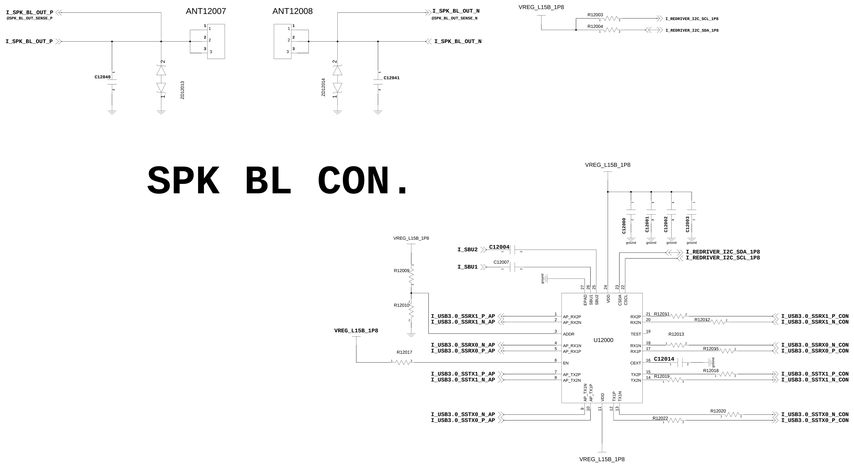


USB PCB	R12006	ADC	Range(mV)	ID
X916B(SUB6)	30K(2007 - 008275)	945.23	394 - 435	1
	100K(2007 - 007107)	2048	683 - 945	3
X910(WIFI)	150K(2007 - 007489)	2057.6	946 - 1134	4
	330K(2007 - 007573)	3143.4	1135 - 1452	5

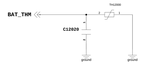


I/F CONN (USB)

SPK BL CON.

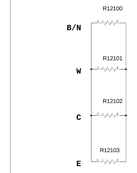


REDRIVER & DP



BAT THM

USB PCB REGION



USB PCB	SHORT
X916(B/N)	R12100
X910(W)	R12101
X916(C)	R12102
ETC(E)	R12103

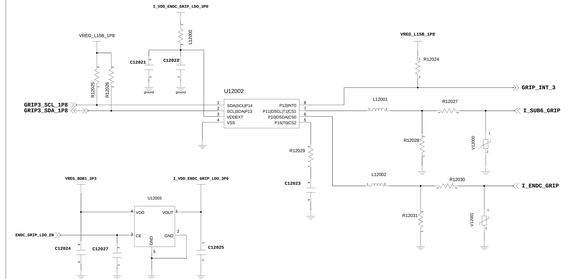
< USB PCB - SMD F/T

1. VREG_L15B_1P8
2. BAT_THM
3. SDM_RFFE1_CLK
4. SDM_RFFE1_DATA
5. GND

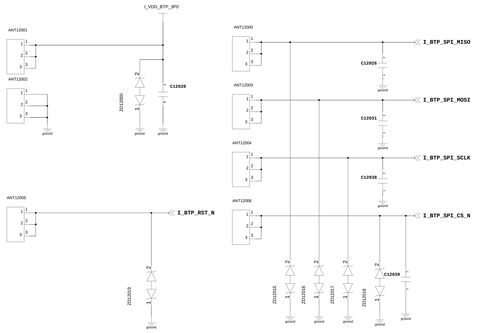
7. VREG_BOB1_3P3
8.
9.
10. ENDC_GRIP_LDO_EN
11. GRIP_INT_3
12. GRIP_SDA_1P8
13. GRIP_SCL_1P8

ENDC_GRIP_SENSOR(GRIP #4,#5)

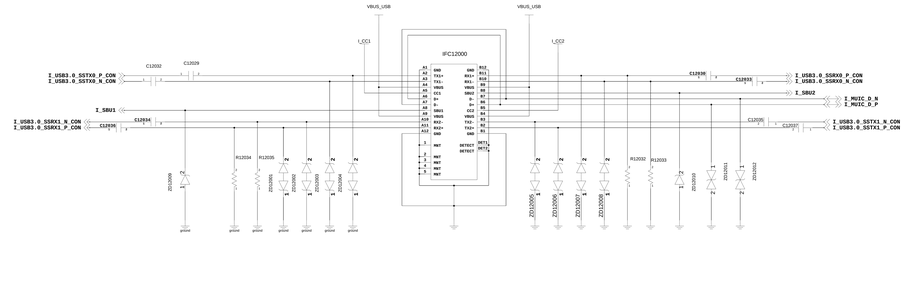
WIFI : NC



ENDC GRIP LDO



FOD CON.

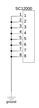


Type-C Conn.

BOTHHOLE/CONTACT

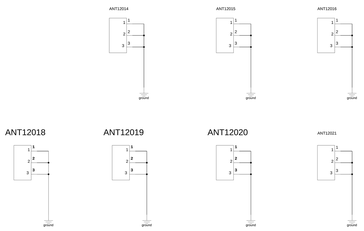
Front CONTACT

SHIELD CAN



Re-Driver

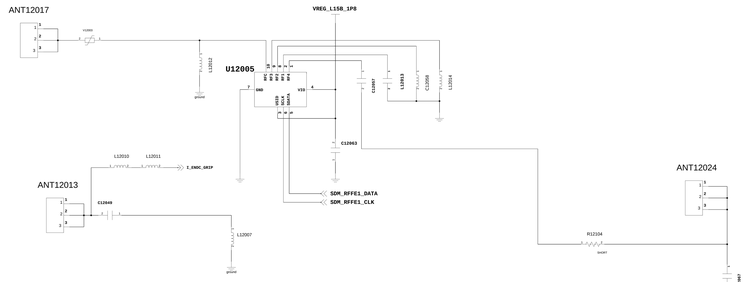
< BOTTOM S/C >



REAR CONTACT(0.9t)

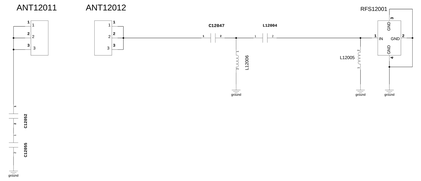
SUB1 ANT

(DRX_LB/MB/HB / DRX_n41 / ENDC_LB/MB TX)



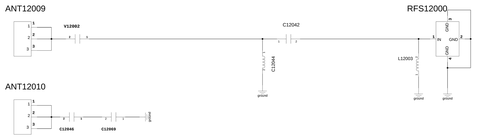
SUB2 ANT

(DRX_MIMO_MB/HB / DRX_n77/n78 / DRX_MIMO_n41)



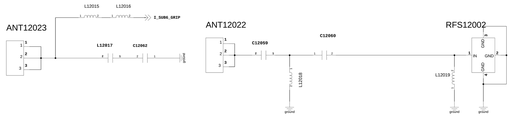
SUB3 ANT

(DRX_MIMO_MB/HB/DRX_MIMO_n41)



SUB4 ANT

(PRX_MIMO_MB/HB / PRX_MIMO_n77/n78 / PRX_MIMO_n41)

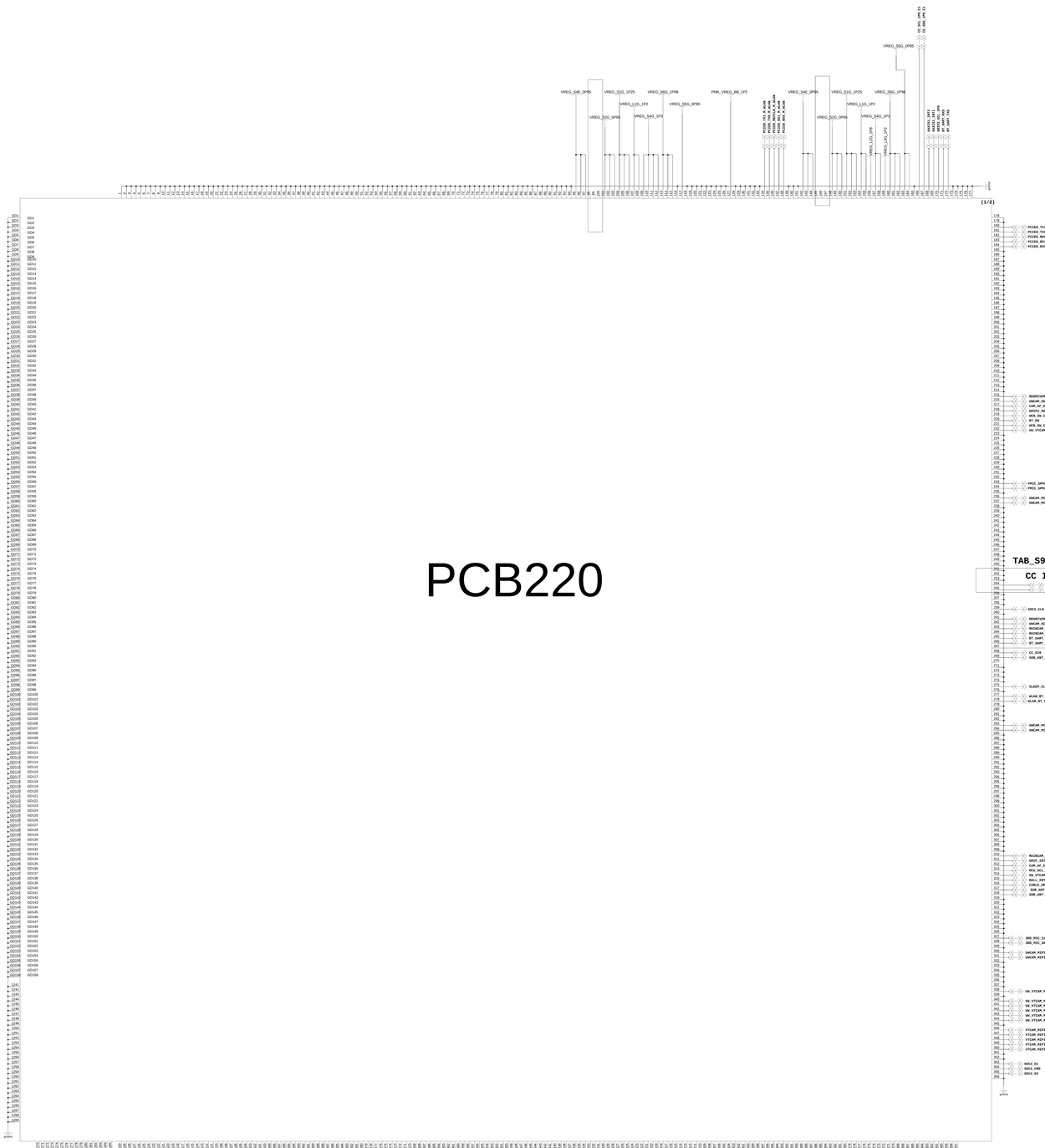


WIFI : NO

Cable CLIP

TITLE: TabS7P				
SIZE: A1	DWG NO: PCB NUMBER			REV:
Drawn/Chk'd: User Name	SHEET: 13	OF: 17	Drawing Date:	

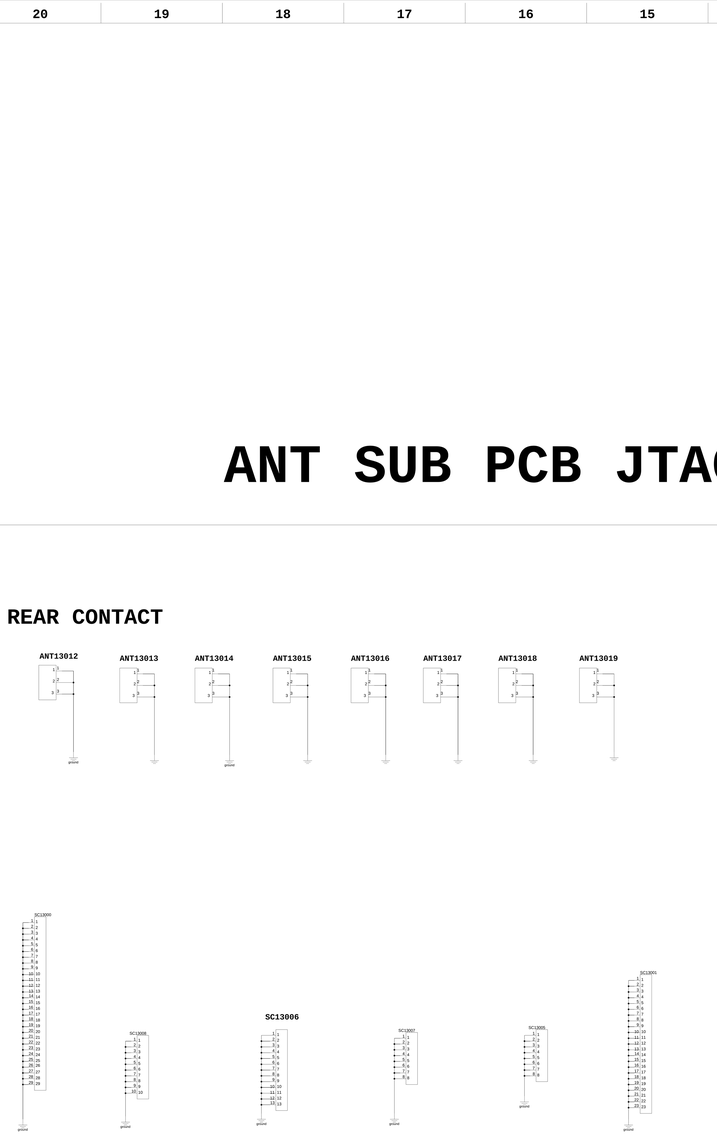
COREPCB PAD


VREG_S4E_0P95
VREG_S1G_1P25
VREG_S5G_1P98
VREG_L1G_1P2
VREG_S4G_1P3
VREG_S2G_0P95
VREG_S5G_0P95
PMK_VREG_BB_1P2
VREG_L3G_1P8
VREG_L3G_1P3
VREG_S4E_0P95
VREG_S1G_1P25
VREG_L1G_1P2
VREG_S4G_1P3
VREG_S2G_0P95
VREG_S5G_1P98
VREG_S5G_0P95
(1/2)
PCIE0_TX1_P_WLAN
PCIE0_TX0_P_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_RX1_P_WLAN
PCIE0_RX0_P_WLAN
SENSOR_SDA_1P8
CAM_AF_EEP_SDA_1P8
SDC2_DAT3
SDC2_DAT2
SDC2_DAT1
SDC2_DAT0
SDC2_CLK
SDC2_CMD
SDC2_D3
SDC2_D2
SDC2_D1
SDC2_D0
SENSOR_SCL_1P8
CAM_AF_EEP_SCL_1P8
SENSOR_MCLK
CAM_MCLK
SENSOR_MIPI_CLK_P
SENSOR_MIPI_CLK_N
SENSOR_MIPI_D0_P
SENSOR_MIPI_D0_N
SENSOR_MIPI_D1_P
SENSOR_MIPI_D1_N
SENSOR_MIPI_D2_P
SENSOR_MIPI_D2_N
SENSOR_MIPI_D3_P
SENSOR_MIPI_D3_N
CAM_RST1_N
CAM_RST2_N
SENSOR_MIPI_CLK_P
SENSOR_MIPI_CLK_N
SENSOR_MIPI_D0_P
SENSOR_MIPI_D0_N
SENSOR_MIPI_D1_P
SENSOR_MIPI_D1_N
SENSOR_MIPI_D2_P
SENSOR_MIPI_D2_N
SENSOR_MIPI_D3_P
SENSOR_MIPI_D3_N
CAM_RST1_N
CAM_RST2_N
CAM_MCLK
CAM_SCL_1P8
CAM_SDA_1P8
PWR_CLK_EXT
PCIE0_CLK_REQ_N
PCIE0_WAKE_N
NC_1P8
SENSOR_MIPI_D3_P
SENSOR_MIPI_D3_N
VM_VTCAM_MCLK
SENSOR_MCLK
CABLE_DET_MCU2
VTM_PWR
PMIC_RB
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_CLK_N
MAINCAM_MCLK
SUBCAM_MCLK
SUBCAM_MIPI_CLK_P
SUBCAM_MIPI_CLK_N
SUB_MCU_SCL
SUB_MCU_SDA
SUBCAM_MIPI_D0_P
SUBCAM_MIPI_D0_N
SUBCAM_MIPI_D1_P
SUBCAM_MIPI_D1_N
SUB_MCU_CLK
SUB_MCU_DATA
PWR_CLK_EXT
SUBCAM_MCLK
SUBCAM_MIPI_D0_P
SUBCAM_MIPI_D0_N
SUB_MCU_CLK
SUB_MCU_DATA
PCIE0_CLK_REQ_N
PCIE0_WAKE_N
NC_1P8
SENSOR_MIPI_D3_P
SENSOR_MIPI_D3_N
VM_VTCAM_MCLK
SUBCAM_MCLK
TAB_S9:ESE_SPI
CC I2C Option
CC_SCL_1P8_I2
CC_SDA_1P8_I2
VREG_L1C_1P1
SLEEP_CLK
WLAN_BT_SLIMBUS_CLK
WLAN_BT_SLIMBUS_DATA
BTFM_RST_N
UWCAM_MIPI_D2_P
UWCAM_MIPI_D2_N
BTFM_SPI_CS_N
BTFM_SPI_CLK
BTFM_SPI_MOSI
BTFM_SPI_MISO
GRIP_INT_1
DDR_MIC_CLK
DDR_MIC_DATA
UWCAM_MIPI_D1_P
UWCAM_MIPI_D1_N
VM_VTCAM_MCLK
VW_VTCAM_MIPI_D1_N
VW_VTCAM_MIPI_CLK_N
VW_VTCAM_MIPI_D0_N
VW_VTCAM_MIPI_CLK_P
VW_VTCAM_MIPI_D1_N
VTCAM_MIPI_CLK_N
VTCAM_MIPI_D0_N
VTCAM_MIPI_D1_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_D0_P
VTCAM_MIPI_D1_P
VW_VTCAM_MIPI_D0_P
VW_VTCAM_MIPI_D1_P
VW_VTCAM_MIPI_CLK_P
VW_VTCAM_MIPI_D1_P
SDC2_D1
SDC2_CMD
SDC2_D2
VTCAM_MCLK
VTCAM_MIPI_D3_P
VTCAM_MIPI_D3_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_D2_N
VTCAM_MIPI_D1_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D0_P
VTCAM_MIPI_D0_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_CLK_N
VW_VTCAM_MIPI_D0_P
VW_VTCAM_MIPI_D0_N
VW_VTCAM_MIPI_CLK_P
VW_VTCAM_MIPI_D1_P
VW_VTCAM_MIPI_D1_N
VW_VTCAM_MIPI_CLK_N
VW_VTCAM_MCLK
VTCAM_MCLK
SDC2_D0
SDC2_CLK
SDC2_D3
SLEEP_CLK
WLAN_BT_SLIMBUS_CLK
WLAN_BT_SLIMBUS_DATA
BTFM_RST_N
BTFM_SPI_CS_N
BTFM_SPI_CLK
BTFM_SPI_MOSI
BTFM_SPI_MISO
UWCAM_MIPI_D2_P
UWCAM_MIPI_D2_N
MAINCAM_SCL_1P8
GRIP_INT_2
CAM_AF_EEP_SCL_1P8
MCU_SCL_1P8
SENSOR_SCL_1P8
CABLE_DET_1P8
SUB_ANT_PCB_ID_2
SUB_ANT_PCB_ID_0

ANT SUB PCB JTAG

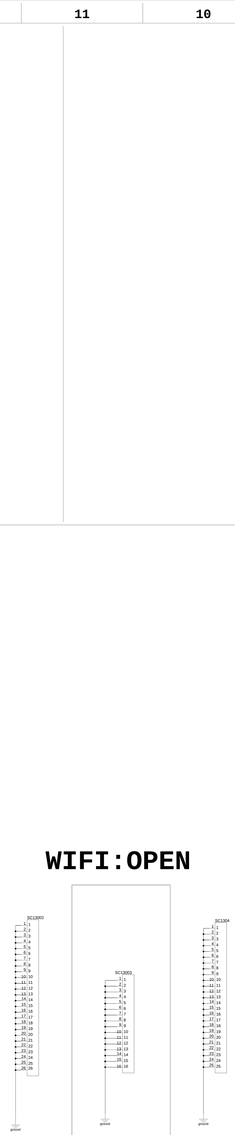
USB SUB PCB JTAG

REAR CONTACT



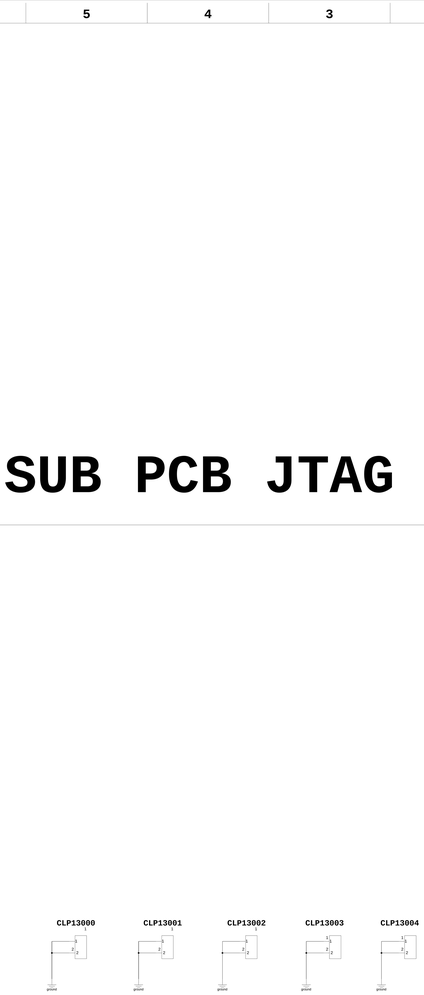
RF1 CLOCK SPK UFS OCTA IF_PMIC

TOP S/C



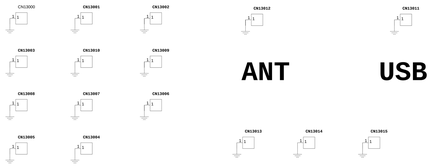
WIFI RF2 AP

BOTTOM S/C



Cable CLIP

Front CONTACT



MAIN

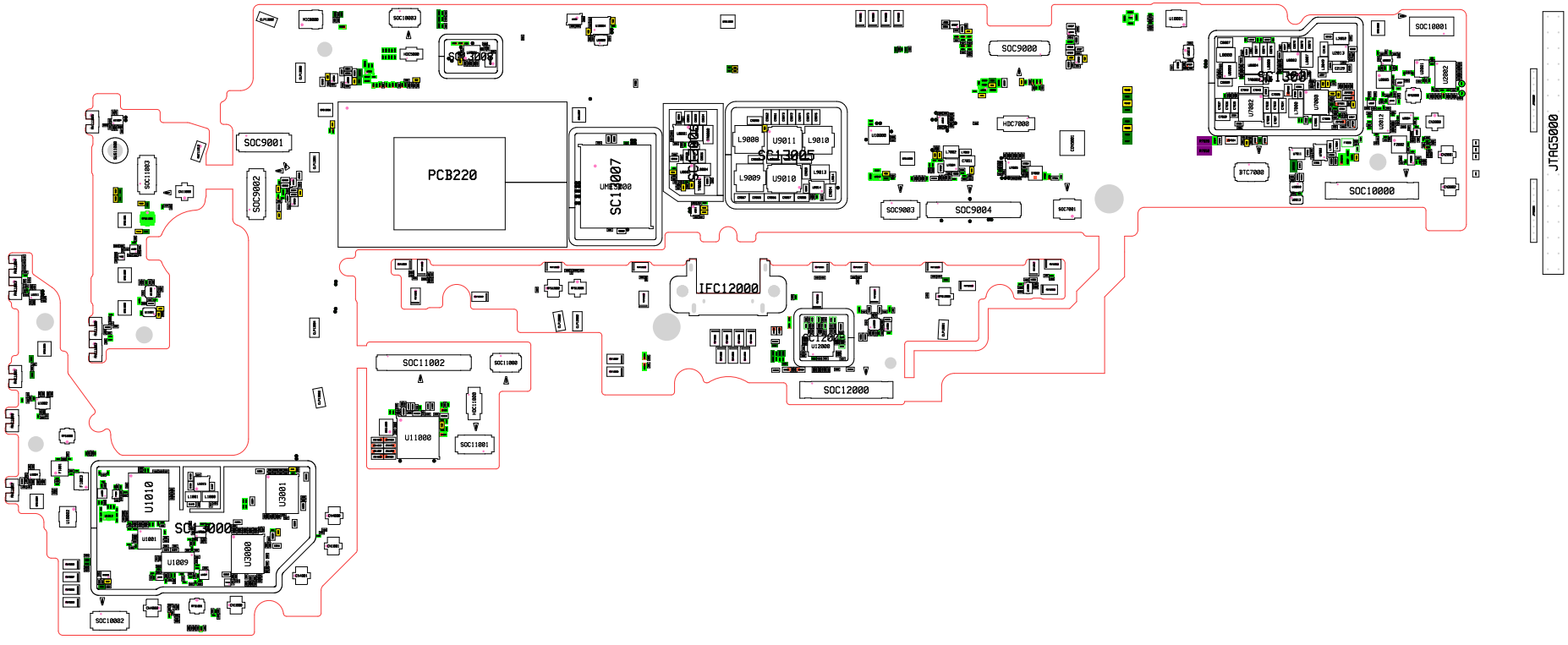
ANT USB

Wacom

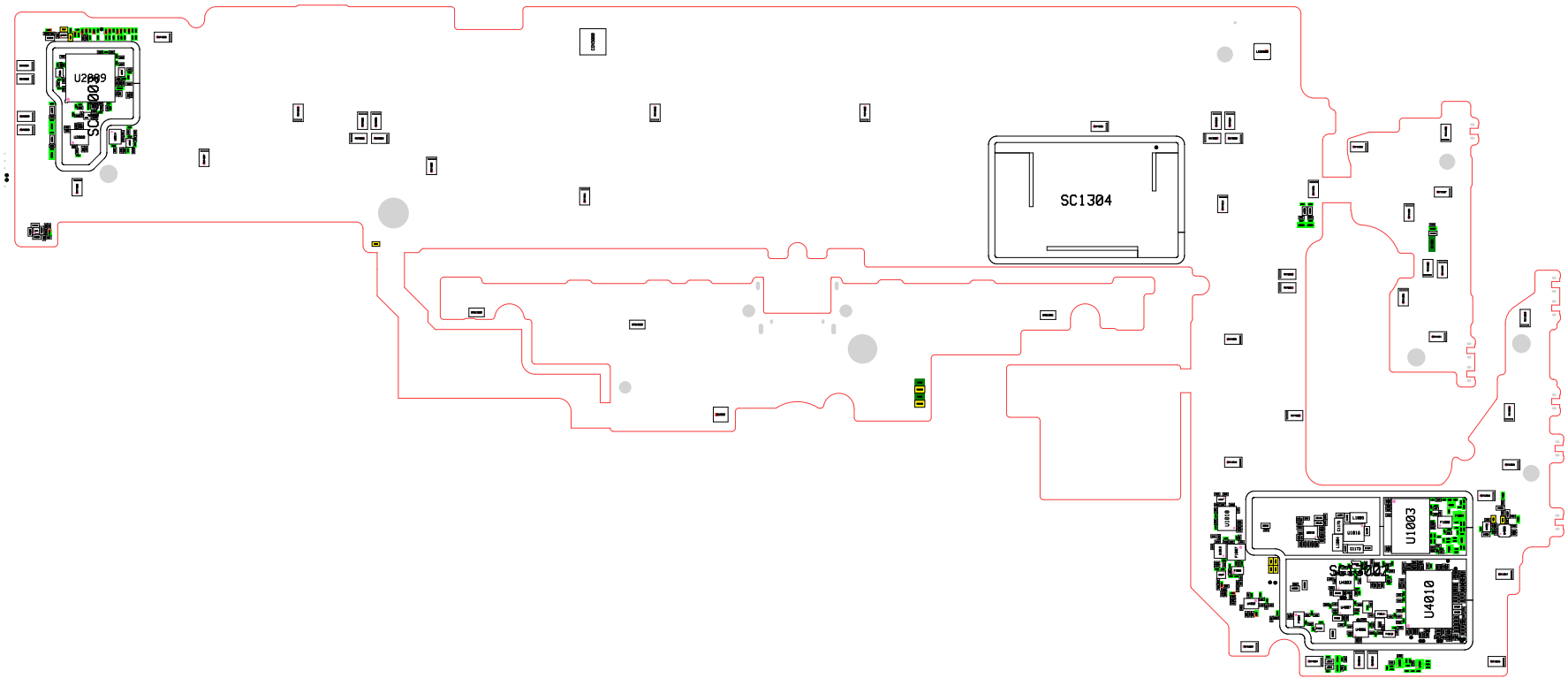
MARK DISASSEMBLY

Title	TabS7P
Size	A1
Number	PDM NUMBER
Rev	
Sheet	13 of 17
Drawing Size	
User Name	

SM-X916B/X916C



SM-X916B/X916C



SM-X910

Service Schematic

TAB_S9_ULTRA_REV0.3

sheet01 : RF (1/2)

sheet02 : RF (2/2), RF PMIC

sheet03 : TRCV (SDR735 x 2)

sheet04 : BT / WIFI(WCN7851), GPS

sheet05 : AP_SM8550(Secondary PCB), JTAG, UFS

sheet06 : AP_SM8550(Secondary PCB)

sheet07 : PMIC(Secondary PCB), IF PMIC(CC IC), D/C

sheet08 : AUDIO

sheet09 : CAMERA / DISPLAY

sheet10 : SENSOR / SD_SIM / KEY / IF

sheet11 : WACOM / ANT SUB PCB

sheet12 : USB PCB

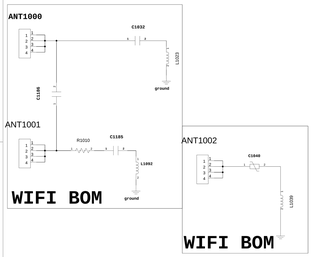
sheet13 : WIFI TR/TL PCB

sheet14 : Secondary PCB PAD

TITLE	TabS7P		
SIZE A1	DWG NO	PDM NUMBER	REV Rev
DRAWN BY User Name	SHEET 0 of 14	Date Drawing Date	

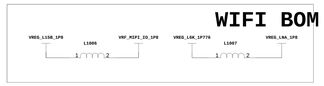
MAIN RF_1

MAIN1 ANT
(TRX_LB/MB/HB / TRX_N41)



MAIN1 ANT XGND1

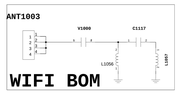
LB LPAMID
Global B5/8/12/13/17/20/26/28, G850/900

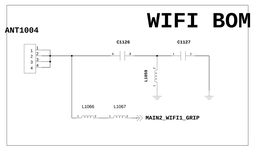


B20RX_B28TRX
EUR : SFX725AUK02 (B20R+B28TR)
NA : SAYRH634MBA0C3K(871 TRX)

2G PAM

B29





SUB5 ANT
(TRX_N77/N78)

close to PA

SFM2215A

UHB PAMID
Global N77/78

OMH LPAMID
Global 1/2/3/4/7/25/32/34/38/39/40/41/66, G1800/1900, N41

THERMISTERS

Global only

NA only

Global only

ET Modulator

LB/UHB/2G ET Modulator for APT (QET5100#0)

SMA point to switching pad

low thickness

MHB/UHB APT Modulator (QET6105#1)

SMA point to switching pad

SUB4 ANT
(PRX_MIMO_MB/HB / PRX_MIMO_n77/78 / PRX_MIMO_n41)

Global : QDM3571
NA : QDM3572

TabS7P

PDM NUMBER

A1

User Name

8 12

Drawing Date

MAIN RF_2

Keep this capacitor

SUB1 ANT
(DRX_LB/MB/HB, DRX_n41, ENDC_LB/MB/HB TX)

SUB2 ANT
(DRX_n77/n78)

DRX_MHB_2
(B2/4/7/25/30/41/66)

Global : QDM3571
NA : QDM3572

SUB3 ANT
(DRX_MIMO_MB/HB, DRX_MIMO_n41, DRX_MIMO_N77/78

DRX_LB
(B5/8/12/13/14/17/20/26/28/29/71_G850/900)

Global : SFMD6B0K001
NA : SFML8A0K001

ENDC module

(B1/2/3/4/7/25/30/39/41/66_DCS/PCS)

THERMISTERS

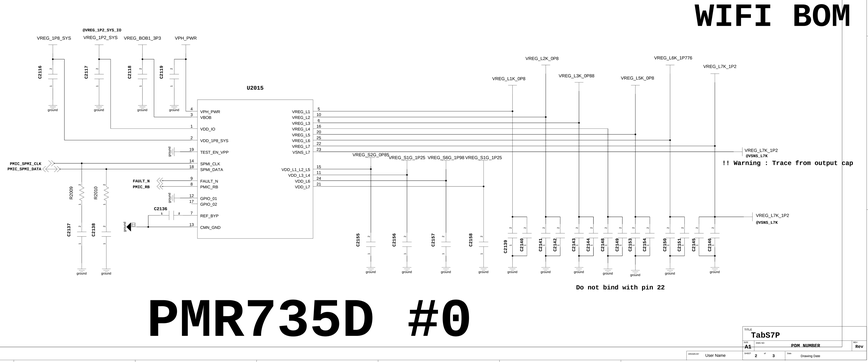
Global(CHN)

ENDC ET Modulator
(QET5100#2)

low thickness

RF PMIC

WIFI BOM



PMR735D #0

TabS7P

A1

FOR_NUMBER

Rev

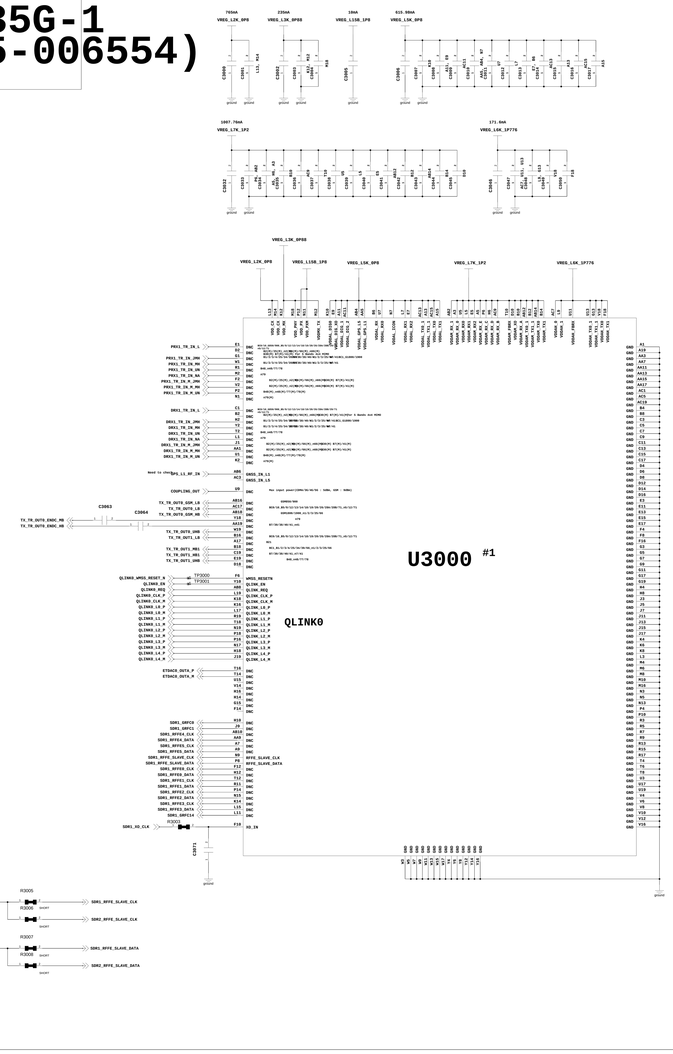
User Name

Drawing Date

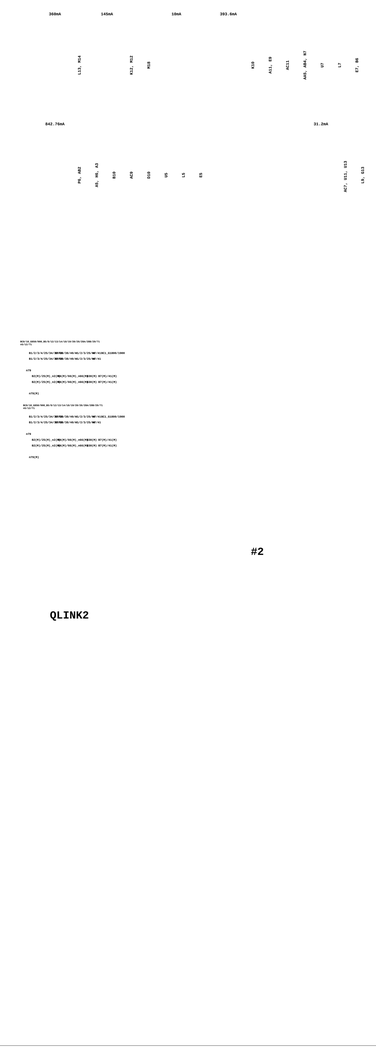
MAIN RF_3

TRANSCEIVER(Dual SDR735)

WIFI : SDR735G-1 (1205-006554)



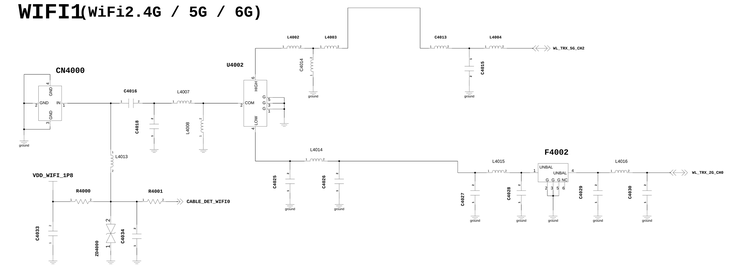
WIFI : NC

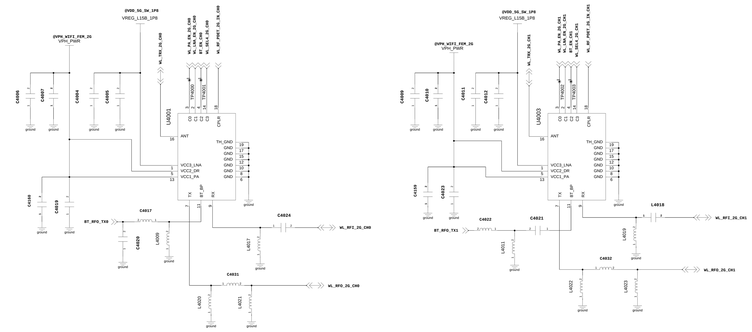


TITLE	TabS7P		
SIZE	DOCUMENT NUMBER		REV
A1	FOR NUMBER		Rev
DRAWN BY	User Name	SHEET 3 OF 16	Drawing Date

Connectivity

WIFI1 (WiFi2.4G / 5G / 6G)



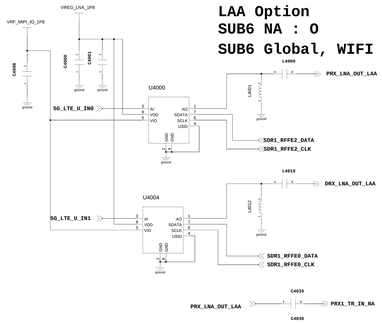


WIFI2 (GPS L1 / WiFi2.4G / 5G / 6G)

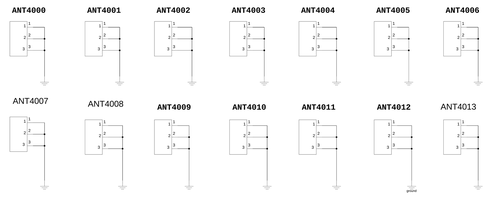
GPS

LAA LNA_Y2

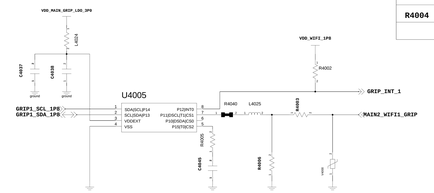
LAA Option
SUB6 NA : O
SUB6 Global, WIFI : X



REAR CONTACT



MAIN_GRIP_SENSOR(GRIP #2_#3)



	Sub6	WIFI
L4027	O	X
R4007	O	X
R4004	O	X

U4010

MAIN GRIP LDO

SM8550
(INTERFACE/GPIO)

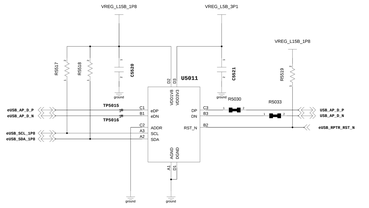
System

CORE

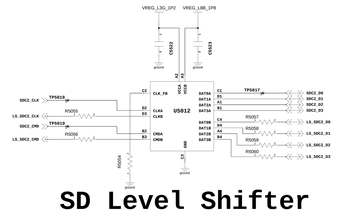
on SECONDARY PCB

WIFI NC

SIM Level Shifter

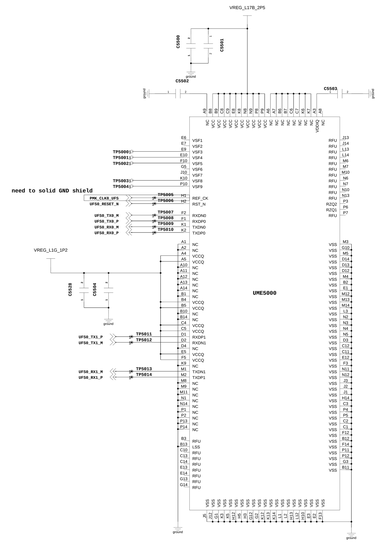


eUSB REPEATER



SD Level Shifter

System

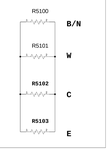


UFS 4.0 256G

0802

REGION

ANT PCB	SHORT
X916(B/N)	R5100
X910(W)	R5101
X916(C)	R5102
ETC (E)	R5103

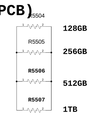


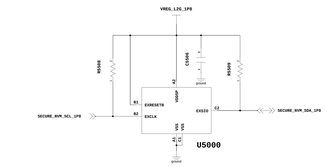
DRAM_POP (S_PCB)

12GB	1105-003212

UFS Memory (M_PCB)

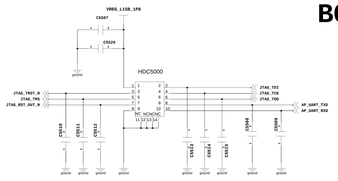
128GB(UFS3.1)	1107- 002751
256GB(UFS4.0)	1107- 002905
512GB(UFS4.0)	1107- 002919
1TB(UFS4.0)	1107- 002924



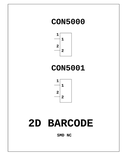


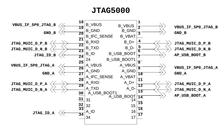
SECURE NVM

BOM NC

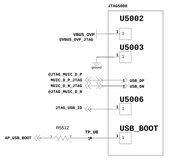


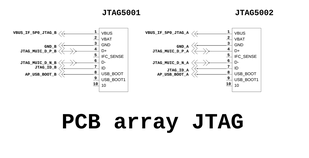
AP JTAG CONNECTOR



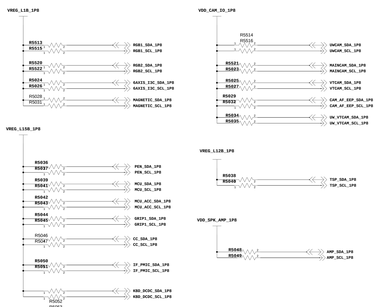


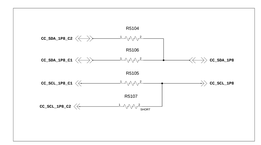
CAUTION: PLACEMENT
Near to TP AP_USB_BOOT





PCB array JTAG





CC I2C Option

TITLE	TabS7P		
SIZE	DOC NUMBER		REV
A1			
SHEET	1	OF	16
DRAWN BY	User Name	Drawing Date	

SM8550 Power

(POWER, GND)

CORE

on SECONDARY PCB

21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1		

TITLE: Tab S7+

SIZE: A1 | DOCUMENT NUMBER: FORM NUMBER | REV: Rev

UNTOUCHED: User Name | SHEET: 6 OF 14 | DATE: Drawing Date

P PMIC set & CLK Generator

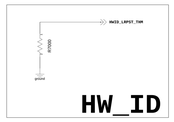
PMIC (PM8550/8550VS/8550VE)

!! CAUTION !!
Place Cin 10uF Bulk Caps at corner of PM8350 and as close to PM8350 as possible to achieve this requirement.

CORE

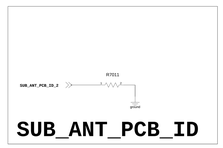
on SECONDARY PCB

Power System



H/W ID table (PM8550 AMUX_4)

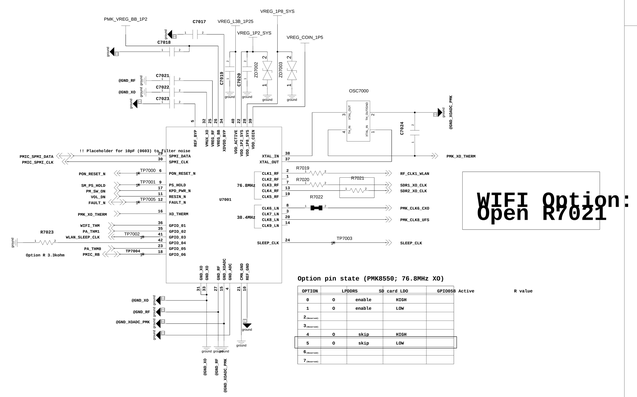
HW ID	BOARD REVISION	Value	Mech
00	Tab S9 Ultra REV0.0	6.04K	(HHP0543)
01		9.1K	
02		12.4K	
03	Tab S9 Ultra REV0.1	18K	2007-007135
04	Tab S9 Ultra REV0.2	22K	2007-007311
05	Tab S9 Ultra REV0.3	30K	2007-008275
06		39K	2007-007134
07		47K	2007-007139
08		56K	
09		68K	2007-007589
10		82K	2007-007590
11		100K	2007-007107
12		120K	2007-000163
13		150K	2007-007489
14		180K	
15		200K	
16			



SUB ANT PBA Table

ID	ID_2	ID_1	ID_0
0	GND	GND	GND Global_Sub6
1	GND	GND	OPEN WIFI
2	GND	OPEN	GND Reserve
3	GND	OPEN	OPEN Reserve

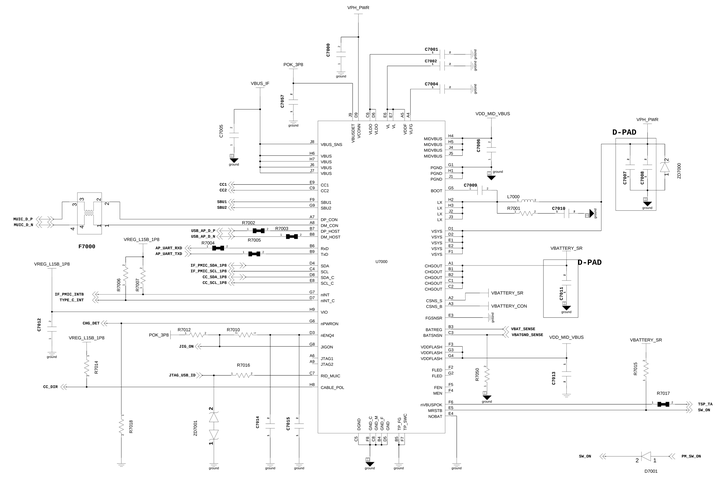
M_RF



WIFI Option:
Open R7021

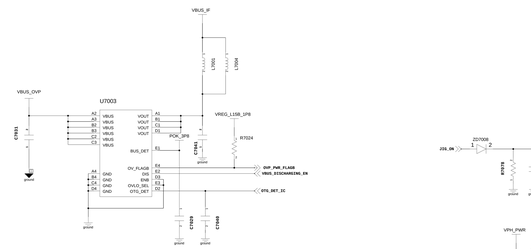
PMK8550

Power

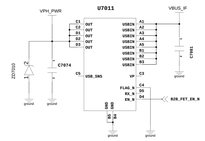


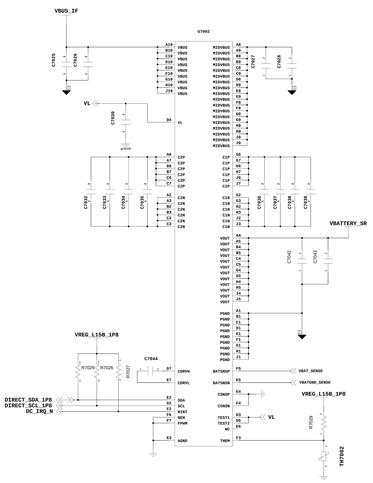
IF PMIC

Power

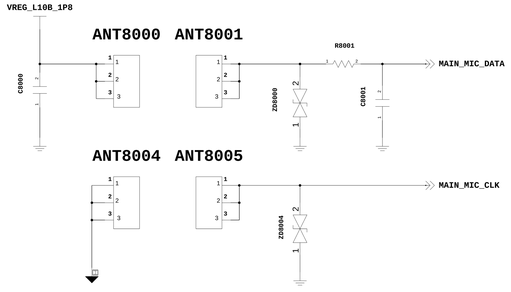


OVP

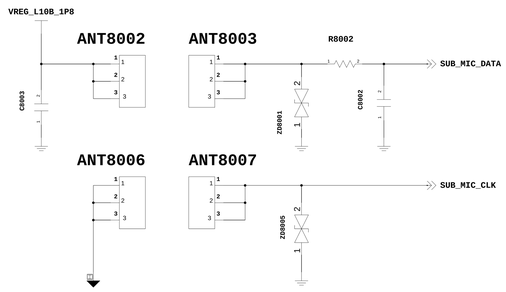




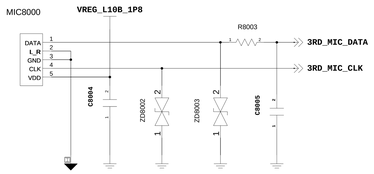
AUDIO



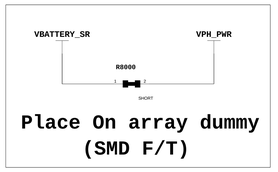
MAIN MIC

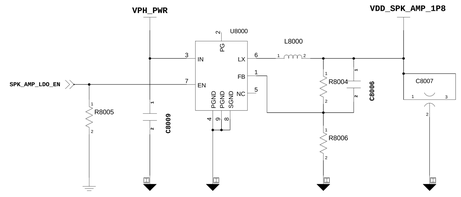


SUB MIC

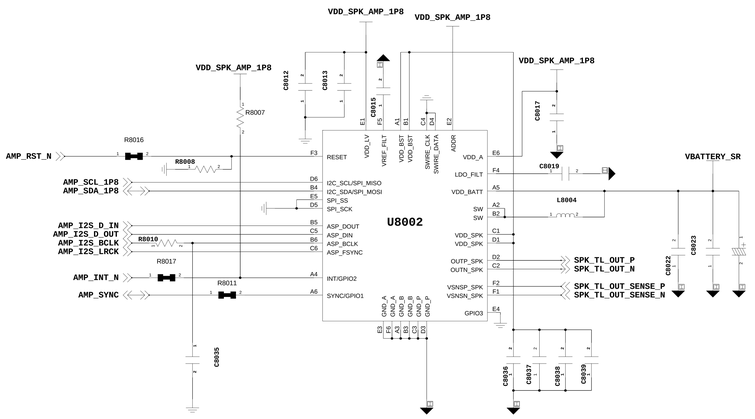


3rd MIC

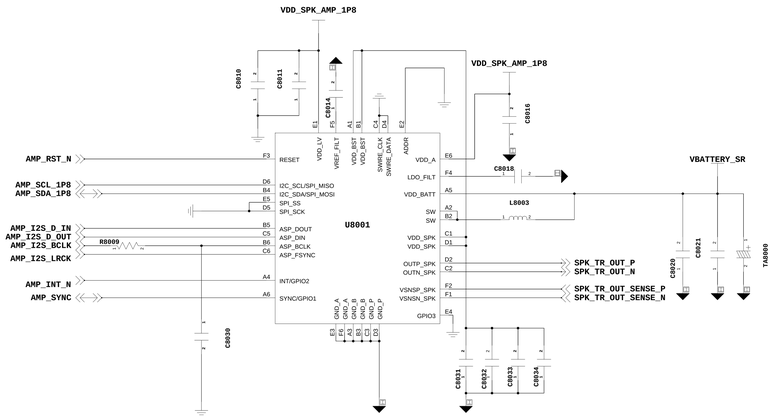




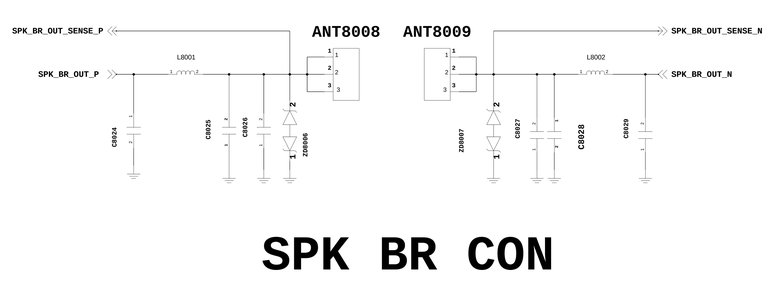
AUDIO DCDC



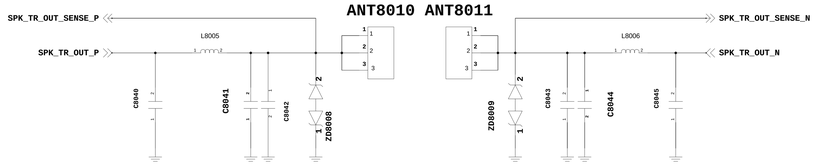
SPK TOP LEFT AMP



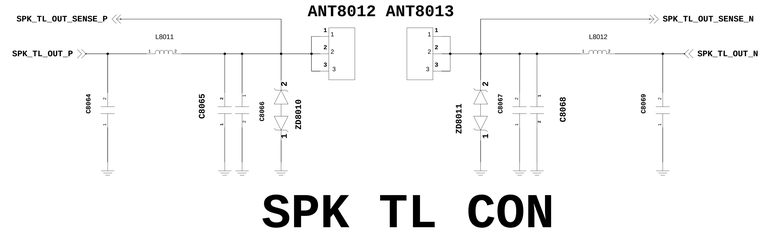
SPK TOP RIGHT AMP



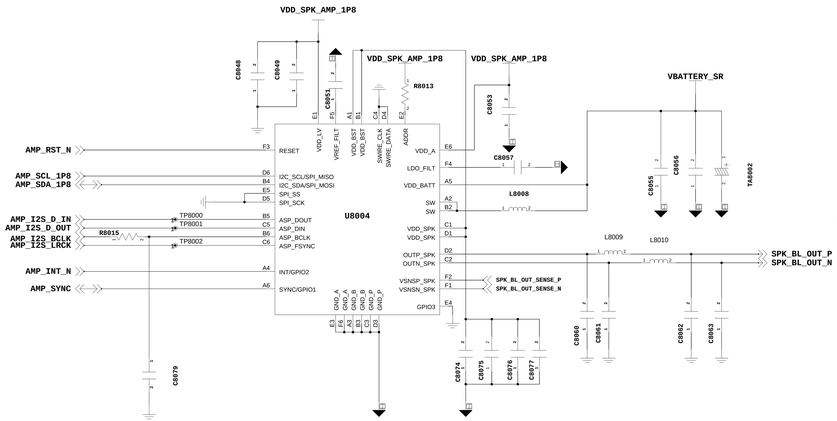
SPK BR CON



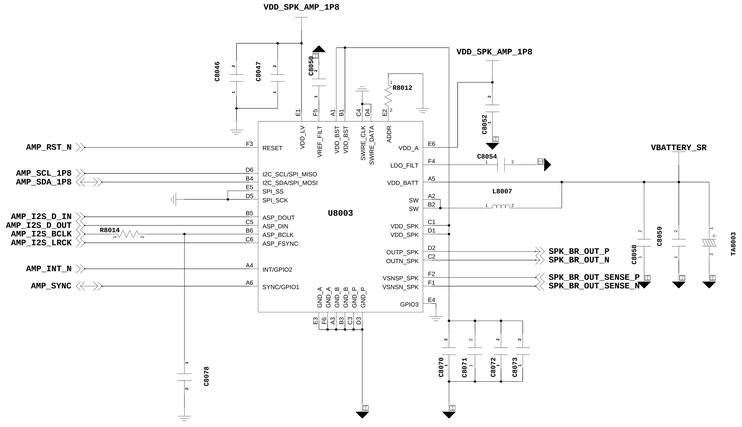
SPK TR CON



SPK TL CON



SPK BOTTOM LEFT AMP



SPK BOTTOM RIGHT AMP

SPK TL CON

TITLE	TabS7P
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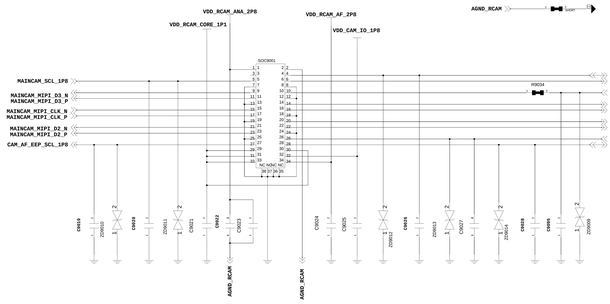
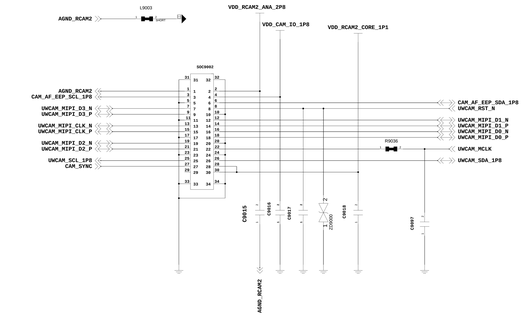
DWG NO	PDM NUMBER
REV	Rev
DRAWN BY	User Name
SHEET	8 of 16
DATE	Drawing Date

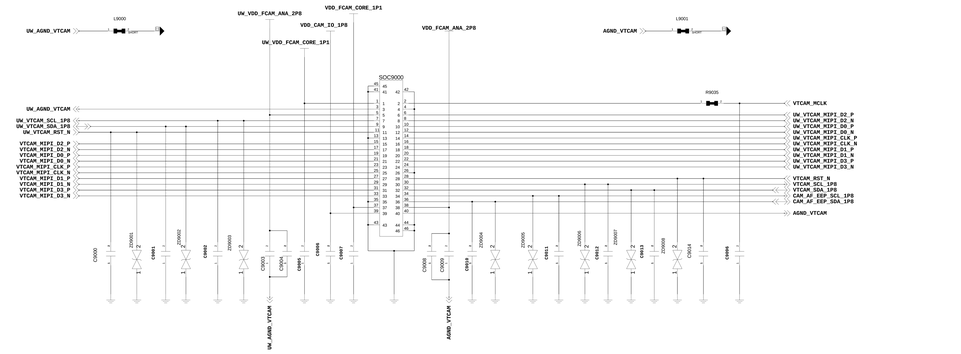
CAMERA



13M_CAM (RCAM)

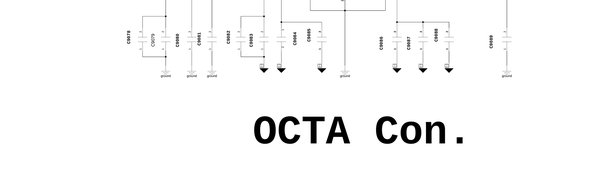


8M_UW CAM (RCAM2)

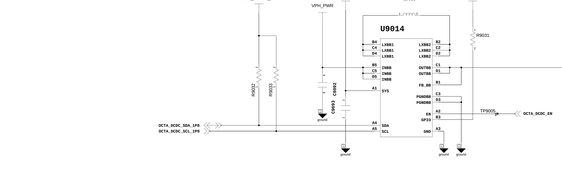


12M CAM (FCAM)

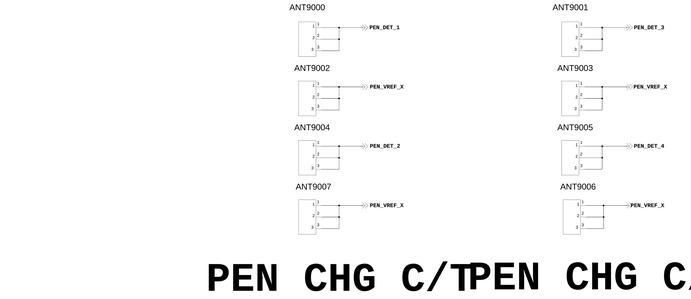
12M_UW CAM (FCAM)



13M_CAM (RCAM)_LDO



8M_UW CAM (RCAM2)_LDO

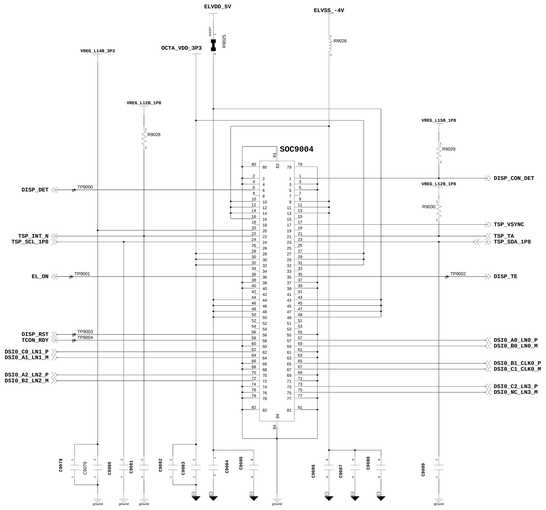


VTCAM_LDO

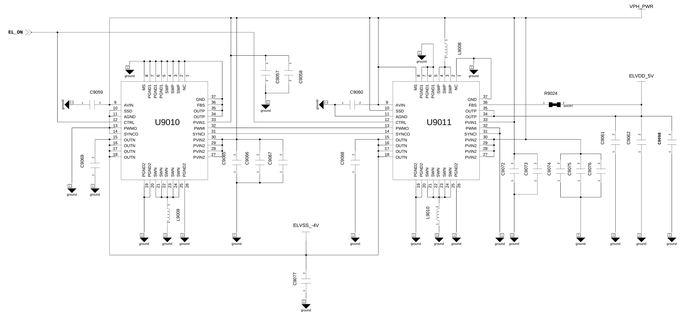
UW VT CAM_LDO

FLASH LED

DISPLAY

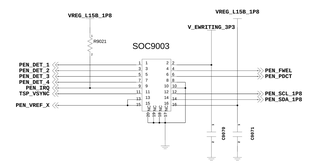


OCTA Con.



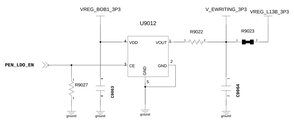
OCTA POWER

TCON BUCK BOOSTER



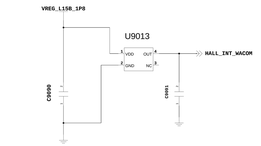
WACOM CONN

(MAIN <-> OCTA / WACOM)



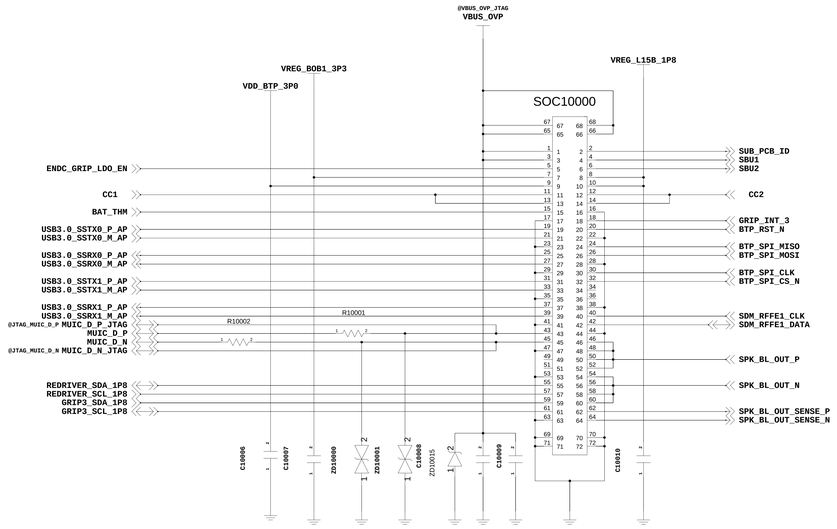
PEN LDO

PEN CHG C/TPEN CHG C/T_2

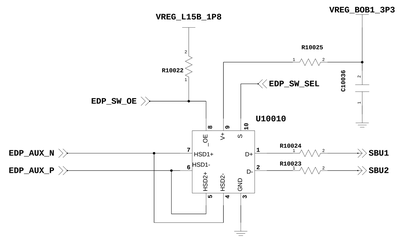


COVER_DET HALL_IC

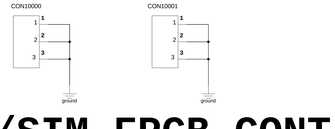
SENSOR / SD/SIM / KEY / I/F



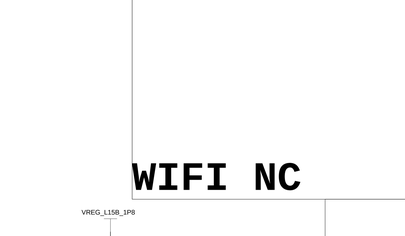
I/F CONN (MAIN <-> USB)



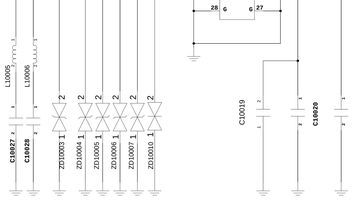
DP Switch



RGB Sensor 1

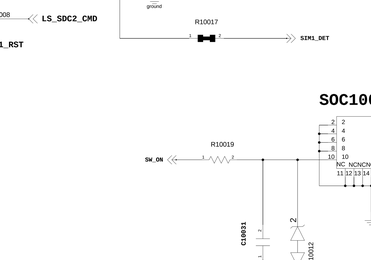


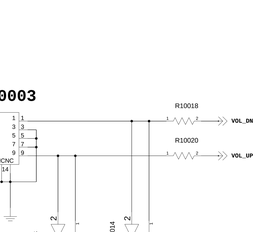
6 Axis (GYRO,ACCEL)



RGB Sensor 2

Option

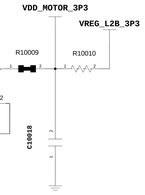




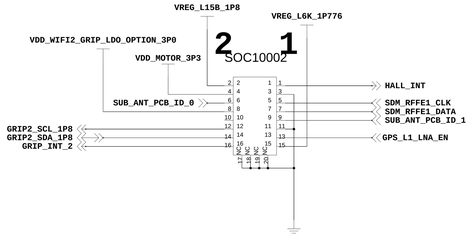
Magnetic SENSOR

SD/SIM FPCB CONTACT

WIFI NC

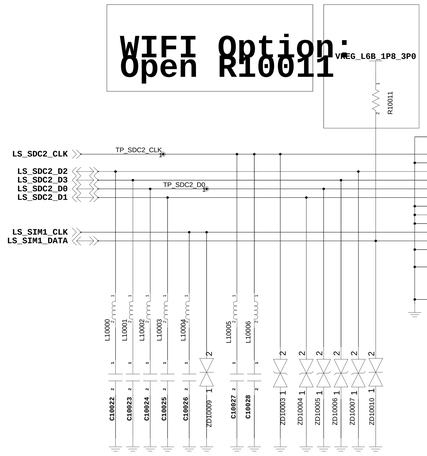


MOTOR LDO



SUB CONN (MAIN <-> ANT SUB)

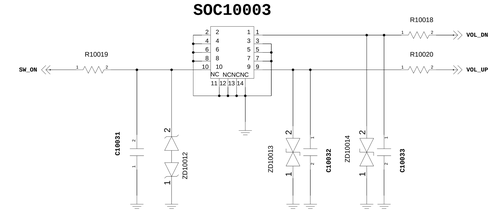
FOD LDO



SIM/SD SOCKET Conn.

WIFI NC

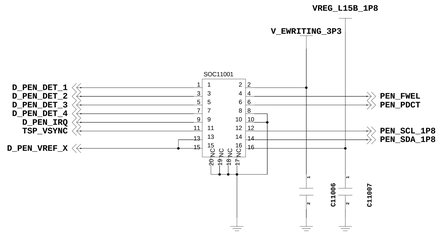
eSIM



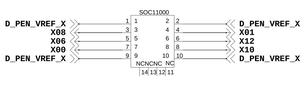
Key CONN

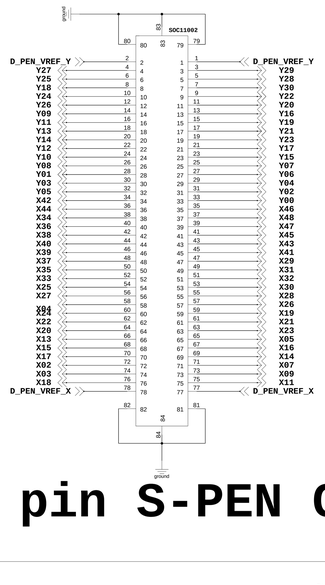
TITLE	TabS7P
SIZE	A1
DWG NO	PDM NUMBER
Rev	
drawn by	User Name
Sheet	10 of 15
Date	Drawing Date

WACOM PCB

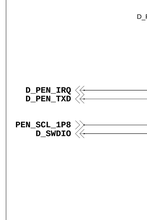


WACOM I/F CON.





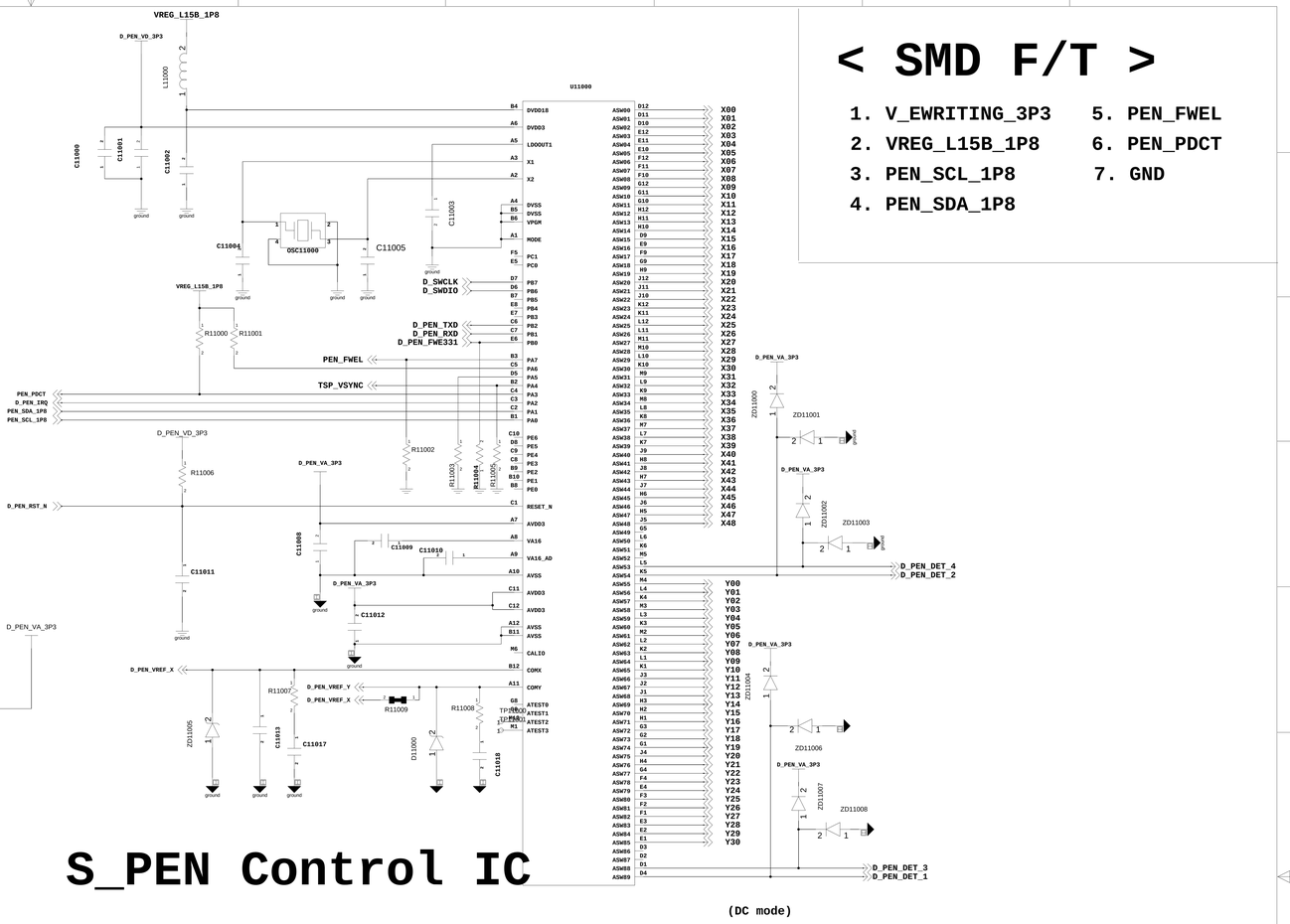
MBOM : NC



DEBUGER

78+10 pin S-PEN Connector

S_PEN Control IC



(DC mode)

< SMD F/T >

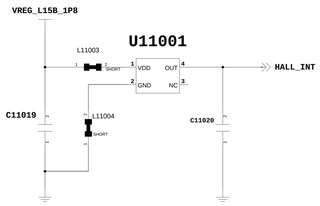
1. V_EWRITING_3P3
2. VREG_L15B_1P8
3. PEN_SCL_1P8
4. PEN_SDA_1P8
5. PEN_FWEL
6. PEN_PDCT
7. GND

ANT SUB PCB

(MAIN1 ANT XGND / WIFI2 ANT GRIP)

SCREW SUS

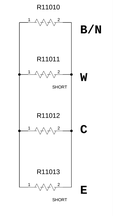
Sub6	WIFI
R11023	R11021
R11024	R11022
	RFS11001



COVER_DET HALL_IC

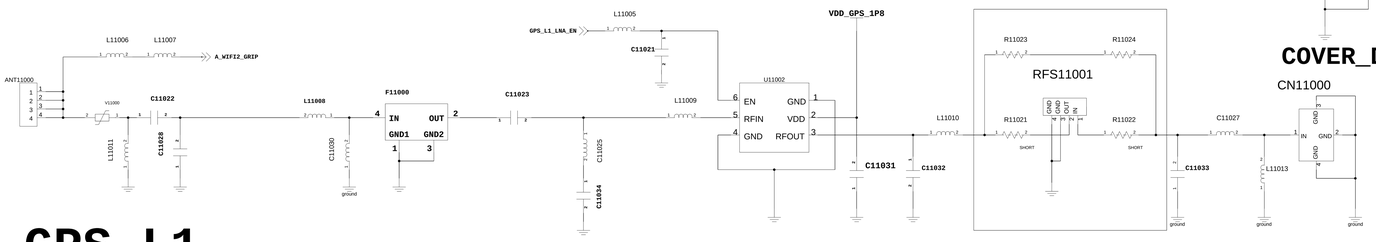
ANT PCB REGION

ANT PCB	SHORT
X916B(B/N)	R11010
X910(W)	R11011
X916(C)	R11012
ETC(E)	R11013

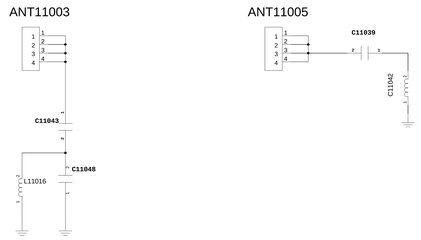


< SMD F/T >

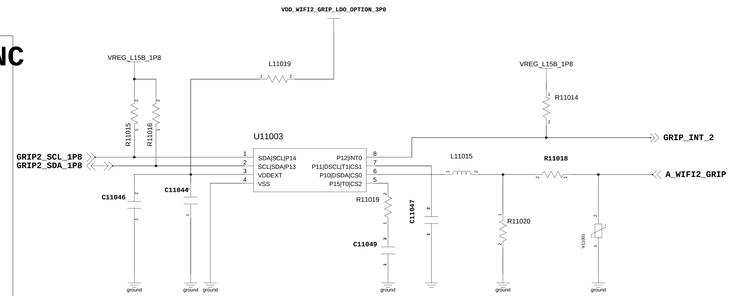
1. VDD_WIFI2_GRIP_OPTION_LDO_3P0
2. GRIP2_SCL_1P8
3. GRIP2_SDA_1P8
4. GRIP_INT_2
5. SDM_RFFE1_CLK
6. SDM_RFFE1_DATA
7. VREG_L15B_1P8
8. HALL_INT
9. GPS_L1_LNA_EN
10. VREG_L6K_1P776
11. GND
12. VPH_PWR
13. GRIP2_LDO_EN



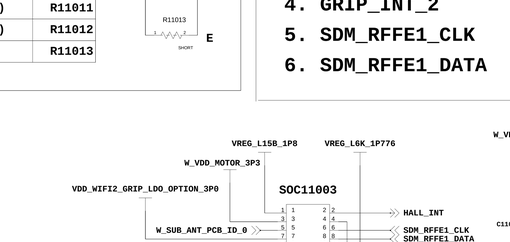
GPS_L1



WIFI : NC

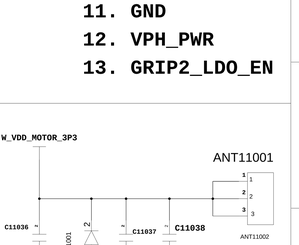


WIFI2_GRIP_SENSOR (GRIP#1)

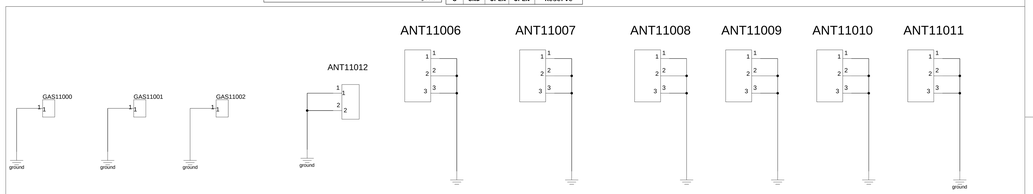


SUB ANT PBA Table

ID	ID_2	ID_1	ID_0
0	GND	GND	GND Global_Sub6
1	GND	GND	OPEN WIFI
2	GND	OPEN	GND Reserve
3	GND	OPEN	OPEN Reserve



MOTOR



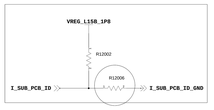
Front CONTACT Cable CLIP REAR CONTACT

TabS7P

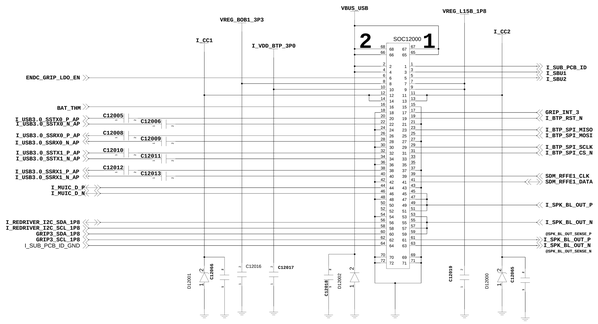
A1 | PDM NUMBER | Rev

User Name | 12 of 14 | Drawing Date

USB PCB

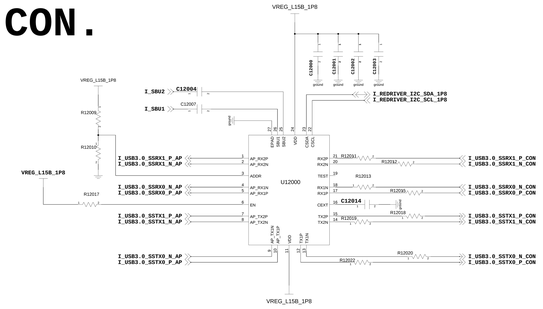


USB PCB	R12006	ADC	Range(mV)	ID
X916B(SUB6)	30K(2007 - 008275)	945.23	394 - 435	1
	100K(2007 - 007107)	2048	683 - 945	3
X910(WIFI)	150K(2007 - 007489)	2057.6	946 - 1134	4
	330K(2007 - 007573)	3143.41	1135 - 1452	5



I/F CONN (USB)

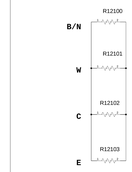
SPK BL CON.



REDRIVER & DP

BAT THM

USB PCB REGION



USB PCB	SHORT
X916(B/N)	R12100
X910(W)	R12101
X916(C)	R12102
ETC(E)	R12103

< USB PCB - SMD F/T

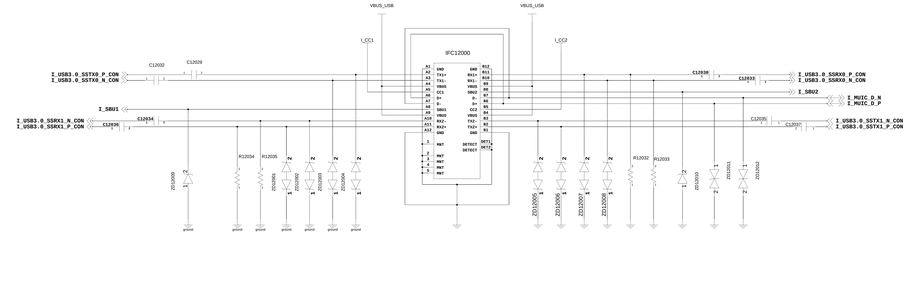
1. VREG_L15B_1P8
2. BAT_THM
3. SDM_RFFE1_CLK
4. SDM_RFFE1_DATA
5. GND

7. VREG_BOB1_3P3
8.
9.
10. ENDC_GRIP_LDO_EN
11. GRIP_INT_3
12. GRIP_SDA_1P8
13. GRIP_SCL_1P8

ENDC_GRIP_SENSOR(GRIP #4,#5)

WIFI : NC

ENDC GRIP LDO

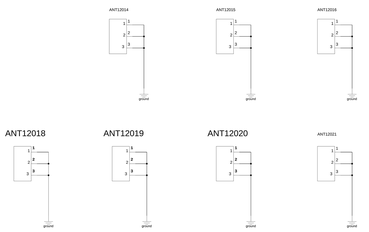


FOD CON.

Type-C Conn.

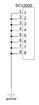
BOTHHOLE/CONTACT

Front CONTACT



REAR CONTACT(0.9t)

SHIELD CAN



Re-Driver

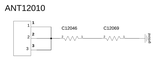
< BOTTOM S/C >

SUB1 ANT

(DRX_LB/MB/HB / DRX_n41 / ENDC_LB/MB TX)

SUB3 ANT

(DRX_MIMO_MB/HB/DRX_MIMO_n41)



SUB2 ANT

(DRX_MIMO_MB/HB / DRX_n77/n78 / DRX_MIMO_n41)

Cable CLIP

SUB4 ANT

(PRX_MIMO_MB/HB / PRX_MIMO_n77/n78 / PRX_MIMO_n41)

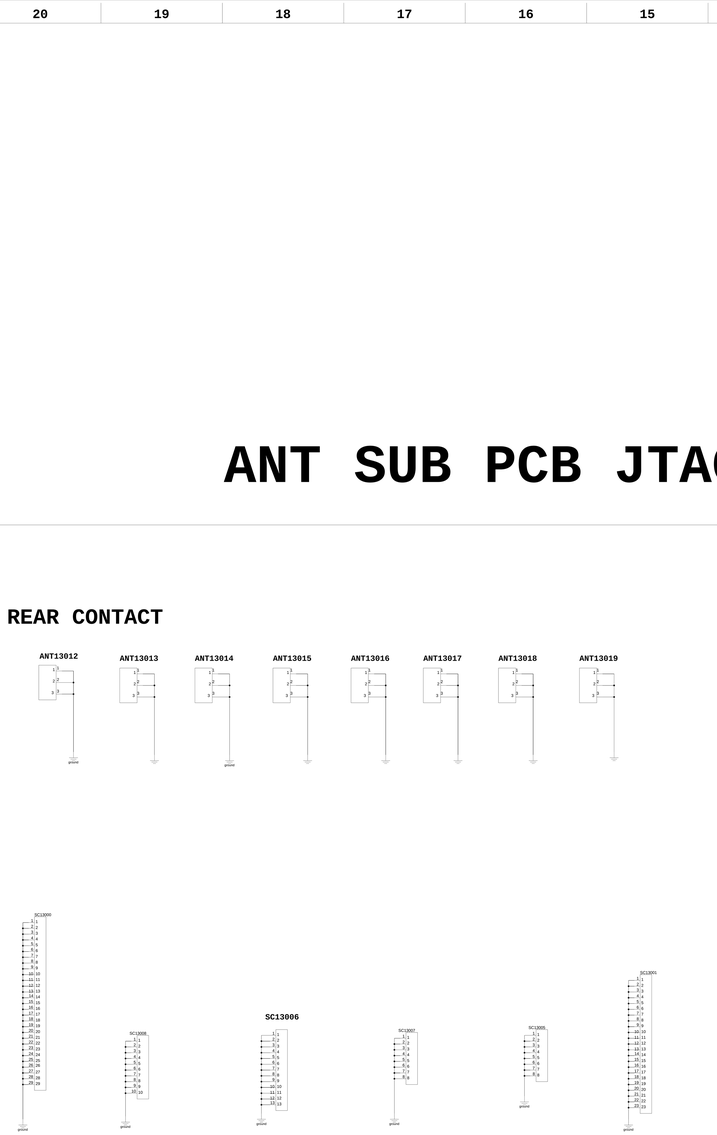
WIFI : NC

Title	TabS7P			
Size	Doc No.	PCB NUMBER		Rev
A1				
Custom	Sheet 13 of 17	Drawing Date		

ANT SUB PCB JTAG

USB SUB PCB JTAG

REAR CONTACT



RF1 CLOCK SPK UFS OCTA IF_PMIC

TOP S/C

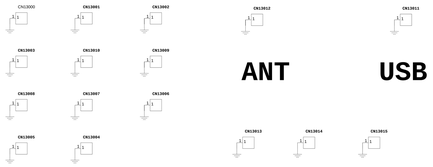
WIFI RF2 AP

BOTTOM S/C

WIFI:OPEN

Cable CLIP

Front CONTACT



MAIN

ANT USB

Wacom

MARK DISASSEMBLY

TITLE	TabS7P
SIZE	A1
DWG NO	PDM NUMBER
REV	Rev
SCALE	13 OF 17
UNIT	Drawing Size
CREATED BY	User Name

COREPCB PAD

PCB220

PCB220

TAB_S9:ESE_SPI
CC I2C Option

BTP_LDO_EN

SSP only(VREG_L3P_1P8)

GRIP2_LDO_EN

CAM_IO_LDO_EN

TabS7P

A1 | PDR NUMBER | Rev

User Name | 22 | 15 | Drawing Date

Tab S9 Ultra Core

Service Schematic

TAB_S9_Ultra_CORE PCB_REV0.5

- CORE PCB TYPE -

2023.04.06

sheet01 : AP_SM8550

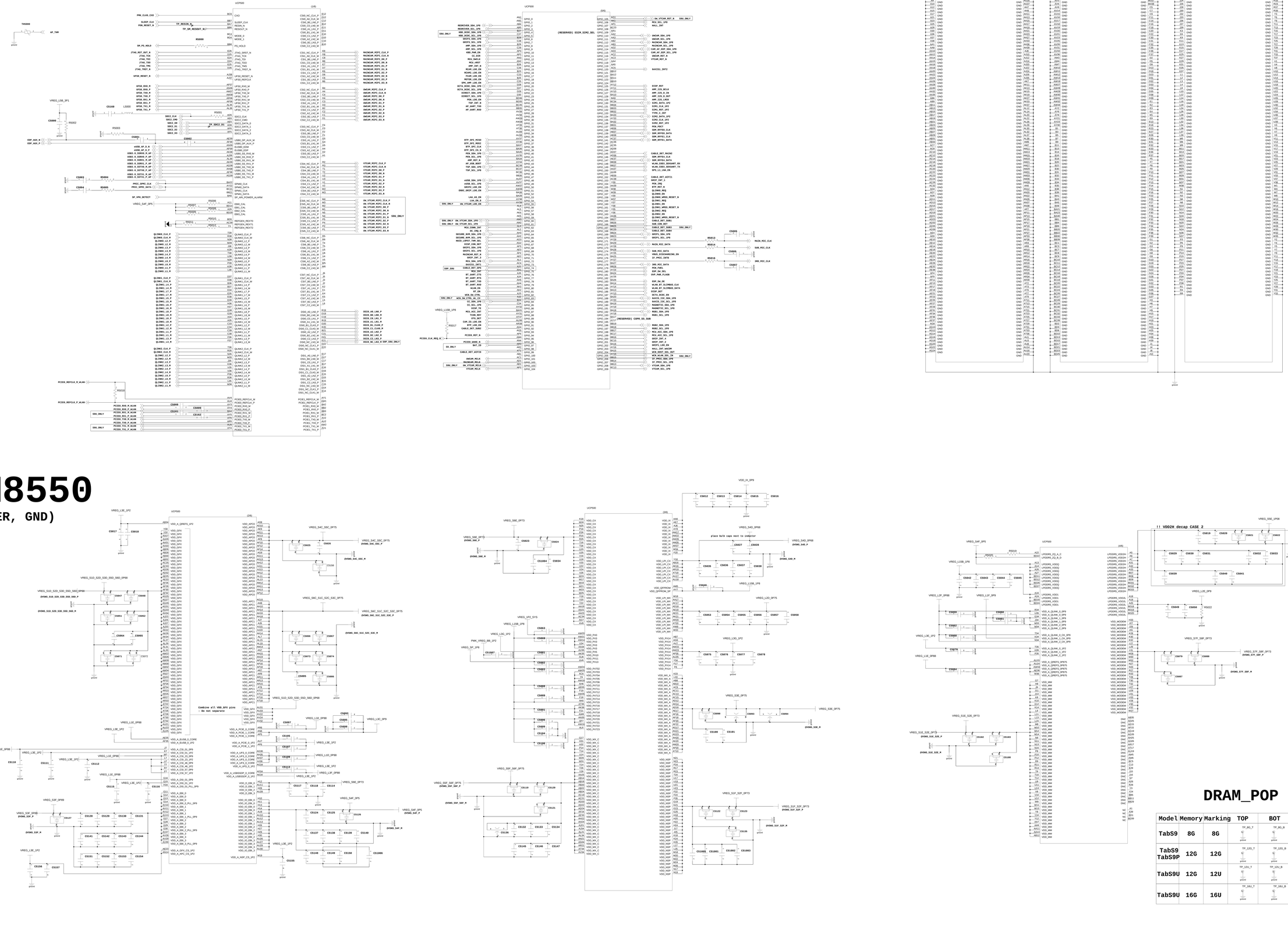
sheet02 : PMIC

sheet03 : CORE PCB

TITLE	TabS9_CORE		
Size: A1	DWG NO: PDM NUMBER		Rev:
DRAWN BY: User Name	Sheet 1 of 4	Date: Drawing Date	

SM8550(INTERFACE/GPIO)(Core PCB) System

mmWAVE, SUB6 : SM8550


UCP500
(1/6)
PMK_CLK48_EXO
SLEEP_CLK
PON_RESET_N
TP_RESIN_N
TP_SM_RESOUT_N
SM_PS_HOLD
JTAG_RST_OUT_N
JTAG_TCK
JTAG_TDI
JTAG_TDO
JTAG_TMS
JTAG_TRST_N
UFS0_RESET_N
UFS0_RXD_M
UFS0_RXD_P
UFS0_TXD_M
UFS0_TXD_P
UFS0_RX1_M
UFS0_RX1_P
UFS0_TX1_M
UFS0_TX1_P
SDC2_CLK
SDC2_CMD
SDC2_D0
SDC2_D1
SDC2_D2
SDC2_D3
SP_SDC2_D1_A2S
SDC4_CLK
SDC4_CMD
SDC4_DATA_0
SDC4_DATA_1
SDC4_DATA_2
SDC4_DATA_3
USB0_DP_AUX_M
USB0_DP_AUX_P
EUSB0_EDM
EUSB0_EDP
USB0_SS_TX1_M
USB0_SS_TX1_P
USB0_SS_RX1_M
USB0_SS_RX1_P
USB0_SS_TX3_M
USB0_SS_TX3_P
SP_VPH_DETECT
VREG_S4F_OPS
EBI_CAL
EBI2_CAL
EBI3_CAL
REFGEN_REXT1
REFGEN_REXT2
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QLINK0_CLK_M
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PCIE0_REFCLK_M_WLAN
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PCIE0_RX0_M_WLAN
PCIE0_RX1_P_WLAN
PCIE0_RX1_M_WLAN
PCIE0_TX0_P_WLAN
PCIE0_TX0_M_WLAN
PCIE0_TX1_P_WLAN
PCIE0_TX1_M_WLAN
(2/6)
CSI0_MC_CLK_P
CSI0_MC_CLK_M
CSI0_MC_LN0_P
CSI0_MC_LN0_M
CSI0_MC_LN1_P
CSI0_MC_LN1_M
CSI0_C3_LN0_P
CSI0_C3_LN0_M
CSI0_C3_LN1_P
CSI0_C3_LN1_M
CSI1_MC_CLK_P
CSI1_MC_CLK_M
CSI1_MC_LN0_P
CSI1_MC_LN0_M
CSI1_MC_LN1_P
CSI1_MC_LN1_M
CSI1_C3_LN0_P
CSI1_C3_LN0_M
CSI1_C3_LN1_P
CSI1_C3_LN1_M
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CSI2_MC_CLK_M
CSI2_MC_LN0_P
CSI2_MC_LN0_M
CSI2_MC_LN1_P
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CSI2_C3_LN1_M
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CSI3_MC_LN1_P
CSI3_MC_LN1_M
CSI3_C3_LN0_P
CSI3_C3_LN0_M
CSI3_C3_LN1_P
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DSI0_MC_CLK_M
DSI0_MC_LN0_P
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DSI0_MC_LN1_P
DSI0_MC_LN1_M
DSI0_C3_LN0_P
DSI0_C3_LN0_M
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DSI1_C3_LN1_M
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PCIE1_REFCLK_M
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PCIE1_RX0_M
PCIE1_RX1_P
PCIE1_RX1_M
PCIE1_TX0_P
PCIE1_TX0_M
PCIE1_TX1_P
PCIE1_TX1_M
(3/6)
GPIO_0
GPIO_1
GPIO_2
GPIO_3
GPIO_4
GPIO_5
GPIO_6
GPIO_7
GPIO_8
GPIO_9
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GPIO_11
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GPIO_99
GPIO_100
GPIO_101
GPIO_102
GPIO_103
GPIO_104
(4/6)
GPIO_105
GPIO_106
GPIO_107
GPIO_108
(RESERVED) ES1R_SIM2_SEL
GPIO_109
GPIO_110
GPIO_111
GPIO_112
GPIO_113
GPIO_114
GPIO_115
GPIO_116
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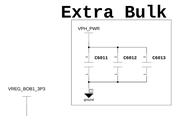
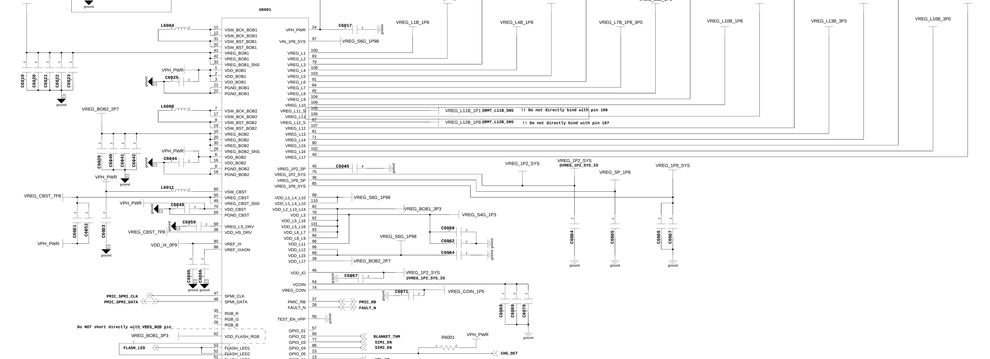
GPIO_184
GPIO_185
GPIO_186
GPIO_187
GPIO_188
GPIO_189
GPIO_190
GPIO_191
GPIO_192
GPIO_193
GPIO_194
(5/6)
IN_VTCAM_RST_N
MCU_SCL_SPU
HALL_INT
SWCAM_SDA_SPU
SWCAM_SCL_SPU
MAINCAM_SDA_SPU
MAINCAM_SCL_SPU
CAM_AF_DIF_SDA_SPU
CAM_AF_DIF_SCL_SPU
SWCAM_RST_N
VTCAM_RST_N
GAS331_INT2
DISP_RST
AMP_I2S_MCLK
AMP_I2S_D_IN
AMP_I2S_LRCK
SIM1_DATA_SPU
SIM1_CLK_SPU
SIM1_RST_SPU
PEN_INT
SIM_RFFE0_CLK
SIM_RFFE0_DATA
SIM_RFFE1_CLK
SIM_RFFE1_DATA
CABLE_DET_MAIN2
SIM_RFFE4_CLK
SIM_RFFE4_DATA
WLAN_CORE_SDIOHWT_RX
WLAN_CORE_SDIOHWT_TX
SPI_LL_LNA_EN
CABLE_DET_WIFI1
GRIP_INT_1
PEN_IRQ
GRIP_RST_N
QLINK3_REQ
QLINK3_WAKE
QLINK3_WMSS_RESET_N
QLINK3_REQ
QLINK3_EN
QLINK3_WMSS_RESET_N
CABLE_DET_SUB1
QLINK2_EN
CABLE_DET_SUB4
SUB_CON_DET
SUB_I2C_SCL_SPU
SUB_I2C_SDA_SPU
MAIN_MIC_DATA
SUB_MIC_DATA
VBUS_ESCHARGING_EN
IF_PMIC_INTB
3RD_MIC_DATA
PEN_PWR1
EDP_ON_DET
EDP_PWR_ENABLE
EDP_ON_SW
WLAN_BT_SLIMBUS_CLK
WLAN_BT_SLIMBUS_DATA
GAS331_INT1
IMA_BIAS_EN
GAS33_I3C_SDA_SPU
GAS33_I3C_SCL_SPU
MAGNETIC_SDA_SPU
MAGNETIC_SCL_SPU
RGB1_SDA_SPU
RGB1_SCL_SPU
RGB2_SDA_SPU
RGB2_SCL_SPU
MCU_ACC_SCL_SPU
MCU_ACC_SDA_SPU
GRIP_INT_2
GRIP_INT_3
GRIPPE_LDO_EN
WCN_WLAN_SDA_3V
WCN_WLAN_SCL_3V
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
VTCAM_SDA_1P8
VTCAM_SCL_1P8
(6/6)
GND
VDD_IR_OPS
VDD_IR
VDD_LPI_CX
VDD_LPI_MX
VDD_LPI_MM
VDD_LPI_MA
VDD_LPI_WX
VDD_LPI_WM
VDD_LPI_AX
VDD_LPI_AM
VDD_QFPROM
VDD_QFPROM_SP
VDD_PX2
VDD_PX3
VDD_PX4
VDD_PX5
VDD_PX6
VDD_PX7
VDD_PX8
VDD_PX9
VDD_PX10
VDD_PX11
VDD_PX12
VDD_PX13
VDD_PX14
VDD_PX15
VDD_PX16
VDD_PX17
VDD_PX18
VDD_PX19
VDD_PX20
VDD_PX21
VDD_PX22
VDD_PX23
VDD_MX_C
VDD_NSP
UCP500
(BA)
UFDQ0_DQ_A_C
UFDQ0_DQ_B_D
UFDQ0_VDDQ
UFDQ0_VDD1
UFDQ0_VDD2

AP PMIC set(Core PCB)

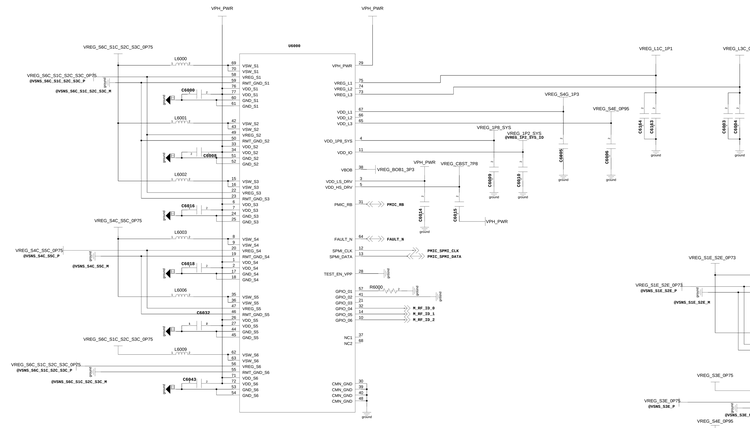
PMIC (PM8550/8550VS/8550VE)

Extra Bulk cap

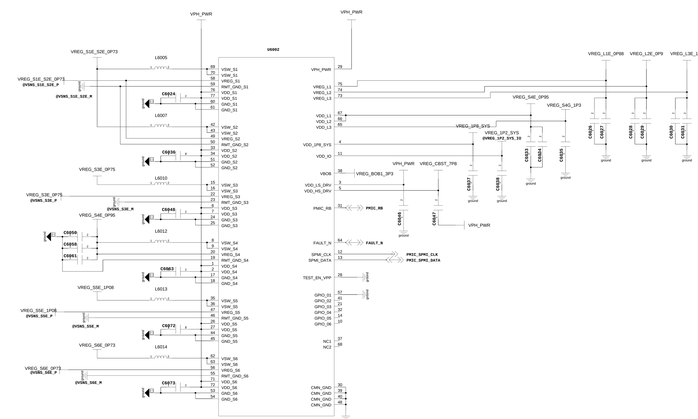




PM8550



PM8550VS_C



PM8550VS_E

PM8550VS_D

PM8550VE_F

PM8550 LDO OUT CAP

TAB S9

HW ID	CORE REV	MAIN REV	Value	Mech
00	REV0.1	REV0.0	6.04K	(HHP0543)
01			9.1K	
02			12.4K	
03			18K	
04			22K	
05			30K	
06			39K	
07			47K	
08			56K	
09			68K	
10			82K	

TAB S9 PLUS

HW ID	CORE REV	MAIN REV	Value	Mech
00	REV0.0	REV0.0	6.04K	(HHP0543)
01	REV0.1	REV0.0	9.1K	
02	REV0.2	REV0.1	12.4K	
03	REV0.2	REV0.2	18K	
04			22K	
05			30K	
06			39K	
07			47K	
08			56K	
09			68K	
10			82K	

TAB S9 ULTRA

HW ID	CORE REV	MAIN REV	Value	Mech
00			6.04K	(HHP0543)
01			9.1K	
02			12.4K	
03			18K	
04			22K	
05			30K	
06			39K	
07			47K	
08			56K	
09			68K	
10			82K	

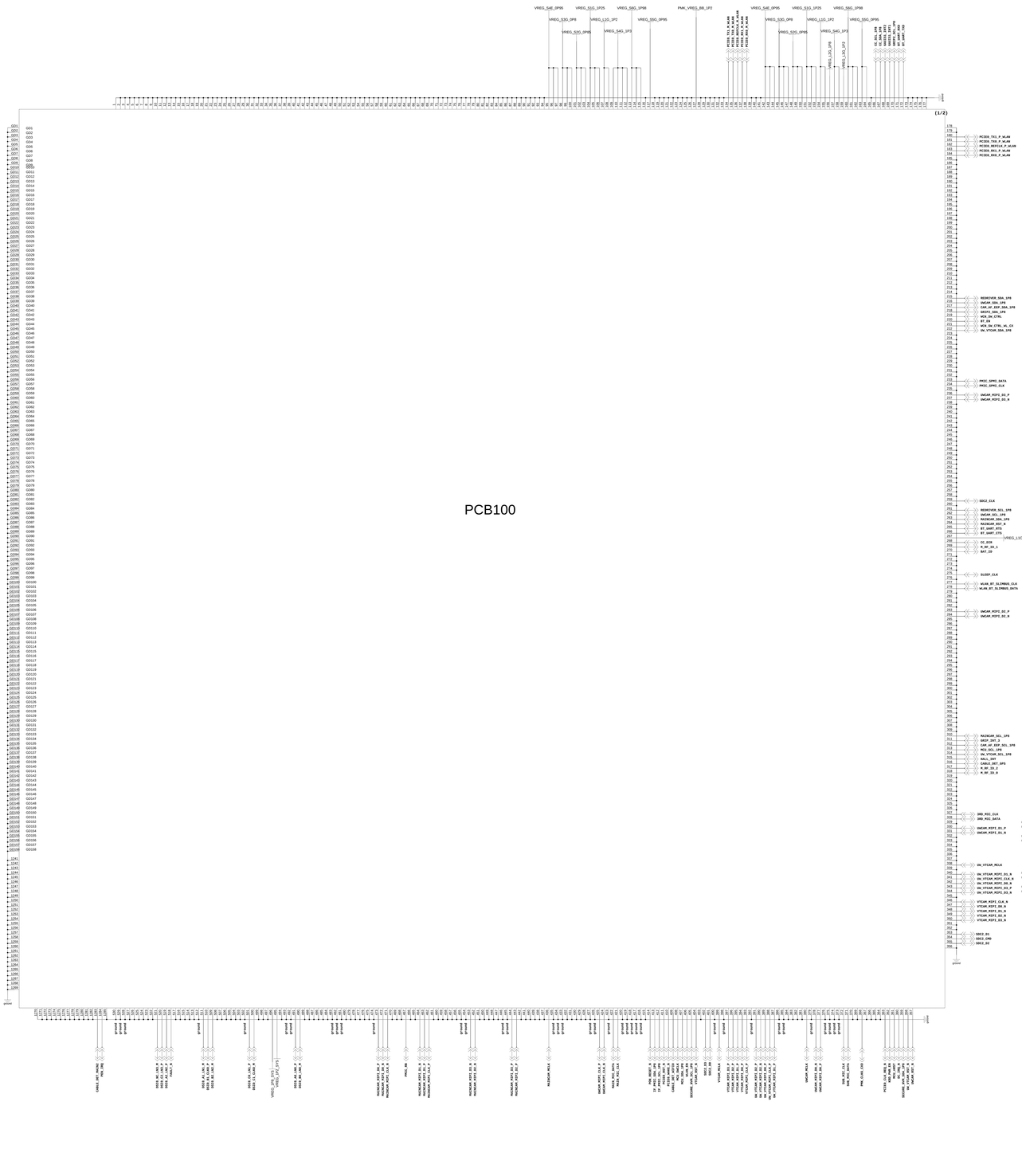
H/W ID table (PM8550 AMUX_4)

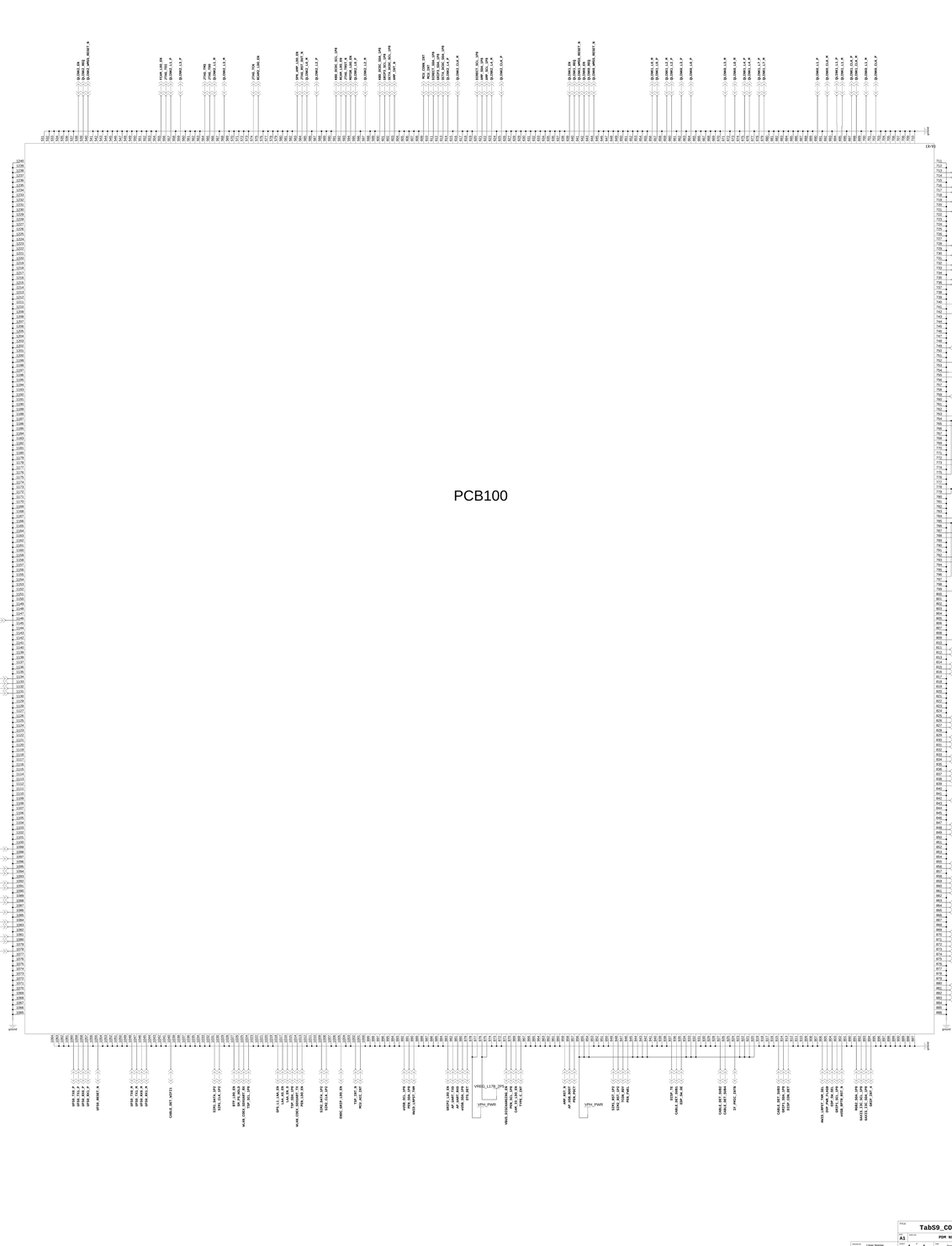
PM8550VS_G

SHIELD CAN 2D BARCORD

TITLE	TabS9_CORE
SIZE	A1
DOC NUMBER	
REV	Rev
SHEET	3 of 4
Drawing Date	
User Name	

COREPCB PAD(Core PCB)





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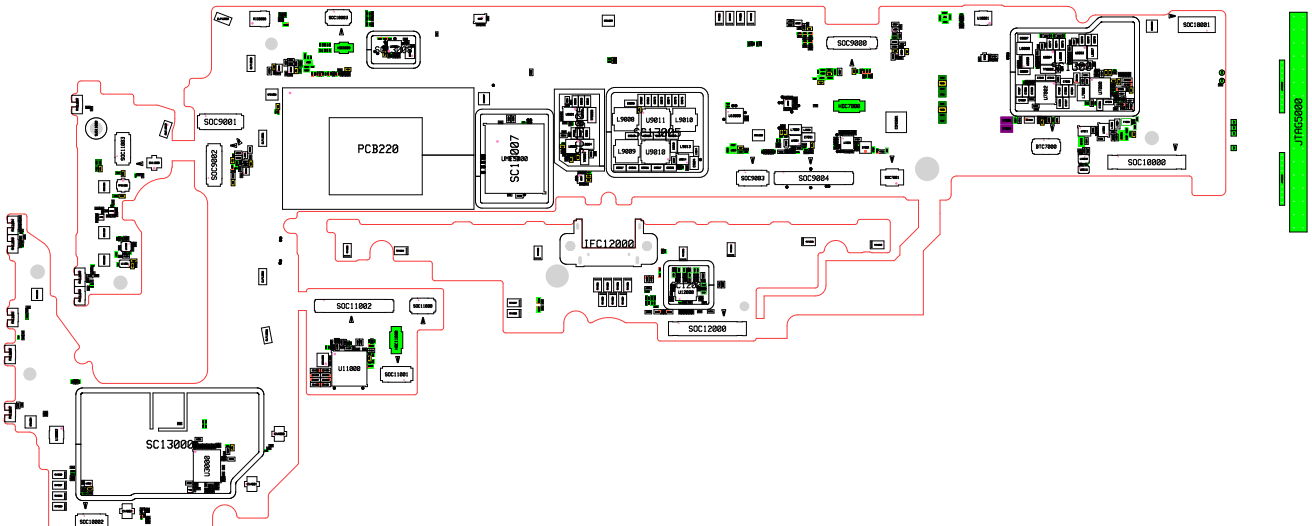
Size	A1
Number	PCM NUMBER
Rev	Rev
Author	User Name
Sheet	4 of 4
Date	Drawing Date

SM-X910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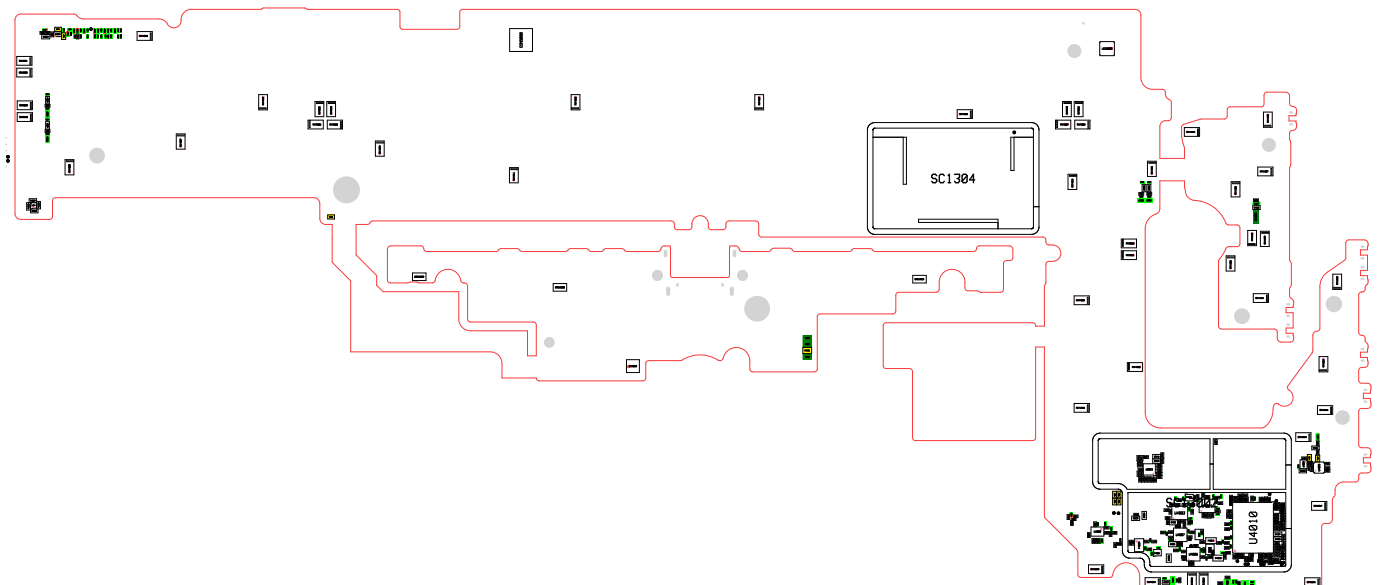
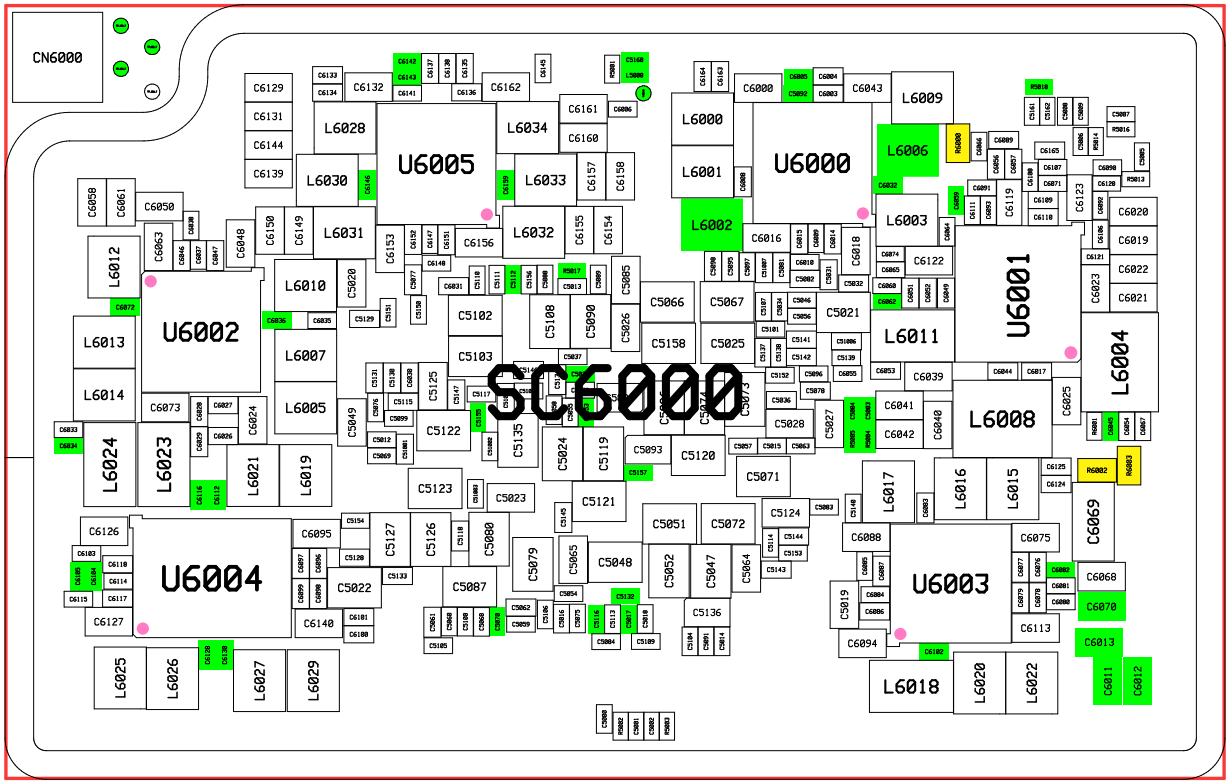
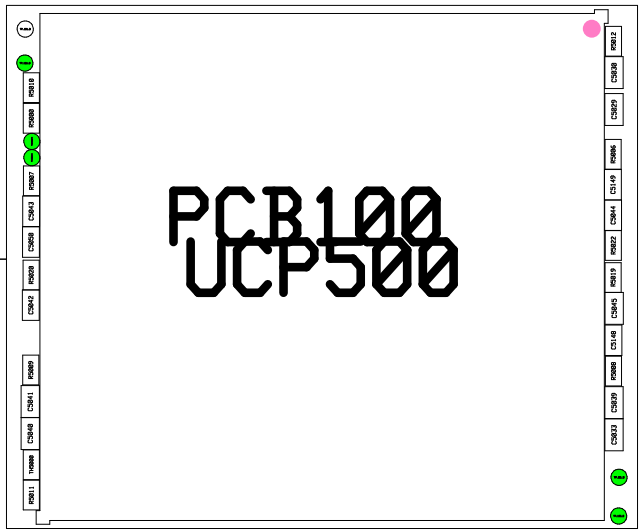
SC13007
SC13005
IFC12000
SC13000
JTAG5000
SOC9001
SOC9002
SOC9080
SOC9083
SOC9084
SOC10001
SOC10000
SOC11002
SOC12000
SOC10002

SM-X910



Tab S9 Ultra Core



Tab S9 Ultra Core